1 (2026) 100010

Contents lists available at ScienceDirect

Bacteriophage

journal homepage: www.sciencedirect.com/journal/bacteriophage

Deciphering the genomic features of four newly isolated polyvalent bacteriophages infecting *Pseudomonas coronafaciens* pv. *garcae*: A tale of small beginnings

Fernanda C. Moreli [a], Carlos A. Quinde [b], Basilio Cieza [c], Aakash Basu [d], Marta M.D.C. Vila [a], Carla Pereira [e], Paul Hyman [f], Victor M. Balcão [a,e,*]

[a] VBlab - Laboratory of Bacterial Viruses, University of Sorocaba, Sorocaba, SP, 18023-000, Brazil
[b] Department of Biological Sciences, University of South Carolina, Columbia, SC, USA
[c] Department of Biophysics and Biophysical Chemistry, Johns Hopkins University, Baltimore, MD, USA
[d] Department of Biosciences, Durham University, Durham, United Kingdom
[e] Department of Biology and CESAM, University of Aveiro, Campus Universitário de Santiago, Aveiro, P-3810-193, Portugal
[f] Department of Biology/Toxicology, Ashland University, Ohio, OH, 44805, USA

ARTICLE INFO

Keywords:
Bacteriophage
Pseudomonas coronafaciens pv. *garcae*
Coffee bacterial blight
Genomic structural features
Genome cyclizability/bendability
Virion morphogenesis yield

ABSTRACT

Emerging phytopathogens are a growing threat to global agriculture, undermining crop yields and jeopardizing food security. Among these, *Pseudomonas coronafaciens* pv. *garcae* (Pcg) causes coffee halo blight, a disease with serious implications for coffee production. Current control measures rely heavily on copper-based compounds and antibiotics like kasugamycin, both of which present environmental hazards and drive resistance development. In this context, bacteriophages offer a targeted and sustainable alternative. Although phage therapy has been explored for several plant pathogens, no phage-based solution has yet been tailored to control Pcg. Here, we describe an in-depth genomic characterization of four novel lytic phages (PCG-05T, PCG-06T, PCG-07T, and PCG-09T) capable of infecting *Pseudomonas coronafaciens* pv. *garcae*. Genomic sequencing and structural analysis via transmission electron microscopy revealed that phages PCG-05T, PCG-07T, and PCG-09T exhibit myovirus morphology, while phage PCG-06T displayed a siphovirus morphotype. Beyond taxonomic classification, we investigated the intrinsic cyclizability of their genomes, integrating insights from DNA biophysics to explore how sequence-dependent mechanical properties influence phage virion assembly. Our findings suggest that DNA bendability, shaped by base pair composition and helical periodicity, may correlate with phage morphogenesis efficiency. This study not only broadens the repertoire of candidate phages for biocontrol of coffee plant pathogens but also introduces an innovative hypothesis linking genome mechanics to phage virion fitness. These insights lay the groundwork for environmentally friendly phage-based strategies to combat bacterial blight in coffee plant cultivation.

1. Introduction

Each year, newly emerging plant pathogens cause significant damage to agricultural systems around the world, resulting in dramatic reductions in crop productivity (Savary et al., 2019). These pathogenic threats compromise not only the volume of harvests but also their commercial value, thereby triggering extensive economic repercussions (Savary et al., 2019). Beyond economic concerns, plant disease epidemics represent a growing danger to global food security and the ecological balance of the planet (Chakraborty and Newton, 2011; Ristaino et al., 2021; Rizzo et al., 2021; Savary and Willocquet, 2020; Singh et al., 2023). Presently, management of bacterial plant infections often involves the application of antimicrobial agents such as the antibiotic kasugamycin or compounds based on copper. However, copper

* Corresponding author. University of Sorocaba (UNISO), VBlab – Laboratory of Bacterial Viruses, Cidade Universitária Prof. Aldo Vannucchi, Rod. Raposo Tavares km 92.5, Sorocaba, SP, São Paulo, CEP 18023-000, Brazil.

E-mail addresses: fernandacamposmoreli@gmail.com (F.C. Moreli), carlosamadeo0708@gmail.com (C.A. Quinde), bciezah1@gmail.com (B. Cieza), aakash.basu@durham.ac.uk (A. Basu), marta.vila@prof.uniso.br (M.M.D.C. Vila), csgp@ua.pt (C. Pereira), phyman@ashland.edu (P. Hyman), victor.balcao@prof.uniso.br (V.M. Balcão).

https://doi.org/10.1016/j.phage.2026.100010
Received 23 February 2026; Received in revised form 28 August 2026; Accepted 29 August 2026

accumulation in the environment presents serious ecological risks and can drive the evolution of resistance in microbial populations (Pinheiro et al., 2019a, 2019b; Yamada, 2014). Likewise, prolonged exposure to kasugamycin can contribute to the selection of resistant phytopathogenic strains (McManus et al., 2002).

The bacterium *Pseudomonas coronafaciens* pv. *garcae* (Pcg) is the etiological agent of coffee halo blight, a disease that threatens coffee cultivation and has drawn increasing attention from the scientific community (Belan et al., 2016; de Souza et al., 2019; Rodrigues et al., 2013, 2015, 2017; Silva et al., 2019, 2023). One promising alternative to synthetic antimicrobials lies in the deployment of bacteriophages—viruses that specifically infect bacteria—as precision tools for targeted biocontrol. These bacteriophages (or phages) offer species-specific antibacterial action, potentially minimizing collateral damage to non-target organisms and the environment. Over recent years, various researchers have explored the use of phages against diverse phytopathogens under laboratory and greenhouse conditions (Frampton et al., 2014, 2012; Harada et al., 2018; Iriarte et al., 2007; Jones et al., 2007; Le, 2019; C. Pereira et al., 2021a,b; Pinheiro et al., 2019a, 2019b; Silva et al., 2023; Svircev et al., 2018; Żaczek et al., 2015; Zaika et al., 2013). Despite such efforts, no phage-based interventions have yet been formulated to counteract the specific threat posed by Pcg in coffee plant crops. Thus, in-depth exploration of the molecular architecture and infection mechanisms of lytic phages targeting Pcg could serve as a foundational step toward environmentally sound and efficient plant disease management strategies.

There is growing scientific interest in understanding how the physical behavior of double-stranded DNA (dsDNA), particularly at shorter sequence lengths, influences its biological functionality. Research suggests that dsDNA exhibits considerable flexibility at spans ranging from 50 to 100 base pairs, contradicting the classical view of DNA as a relatively rigid molecule (Basu et al., 2021a, 2021b; Dickerson, 1998; Tang, 2021). This intrinsic flexibility (commonly referred to as "cyclizability") is vital to numerous genomic processes (Alexandrov et al., 2016; Balcão et al., 2022; Vafabakhsh and Ha, 2012), and is strongly dependent on the underlying base pair composition (Balcão et al., 2022; Basu et al., 2021a, 2021b; Harada et al., 2022). Sequences characterized by clusters of A/T dinucleotides flanked by G/C-rich regions, especially when those A/T stretches align with the natural helical periodicity of DNA, exhibit elevated cyclizability. Such mechanical features of DNA may exert significant influence over transcriptional dynamics, as genome looping and enhancer-promoter proximity in prokaryotic systems are tightly governed by DNA flexibility (Basu et al., 2021a, 2021b). Conversely, segments of DNA with low bendability (typically enriched in G/C content) display reduced cyclizability, which might impede efficient engagement with transcriptional machinery. DNA sequences with greater bendability, particularly those rich in A/T on the outer face of the double helix, facilitate more favorable interactions with RNA polymerases (Balcão et al., 2022; Basu et al., 2021a, 2021b).

Protein-DNA interactions are fundamental to a wide range of regulatory mechanisms within cells, and these interactions are heavily influenced by the mechanical and structural characteristics of the DNA molecule itself (Rohs et al., 2009). The ability of DNA to undergo conformational changes, such as bending, twisting, and deformation, is essential for enabling or restricting access to regulatory proteins. These dynamic behaviors are closely tied to the DNA's three-dimensional shape and local structural parameters (Abe et al., 2015), which include factors like minor groove width (MGW), helical twist (HelT), propeller twist (ProT), and roll angles.

Helical twist (HelT) refers to the angle of rotation needed to align one base pair with the next along the helical axis of the DNA strand (Dickerson and Klug, 1983; Kriegel et al., 2018). The geometry of the minor groove, in particular, is a critical element recognized by many DNA-binding proteins and plays a key role in molecular recognition and binding specificity (Laughton and Luisi, 1999; Stella et al., 2010). Propeller twist (ProT) describes the torsional deformation around the long axis of the base pair, often resulting in a non-coplanar arrangement of the bases within the pair (El Hassan and Calladine, 1996). Meanwhile, the roll angle represents the degree to which adjacent base pairs tilt toward or away from the major or minor grooves, contributing to localized DNA curvature (Tolstorukov et al., 2007).

Together, these parameters define the physical pliability (or bendability) of DNA, shaping the landscape of protein accessibility and interaction dynamics at a molecular level. Understanding these nuances sheds light on how DNA structure influences its biological functionality and the regulation of genetic information.

Genome mechanical features can now be predicted based on the DNA sequence, via using high-throughput methods, revealing inherent preferred conformations (Ussery, 2002; Zhou et al., 2015).

The hypothesis explored here is novel in the context of phage biology: by correlating the mechanical behavior of phage DNA with the efficiency of virion assembly inside host bacteria, new insights may emerge regarding viral replication success.

In the present study, we report the full genomic characterization of four newly identified lytic bacteriophages that infect *Pseudomonas coronafaciens* pv. *garcae*, a taxonomic variant closely related to *Pseudomonas syringae* pv. *garcae*. These phages (designated PCG-05T, PCG-06T, PCG-07T, and PCG-09T) genomes underwent a comprehensive structural and molecular analysis. In addition to sequencing and functional annotation of their genomes, we also examined the intrinsic cyclizability and mechanical properties of their dsDNA sequences to investigate potential connections between DNA bendability and viral morphogenesis yields.

2. Materials and methods

2.1. Biological material

The four phages described in this work (PCG-05T, PCG-06T, PCG-07T and PCG-09T) were isolated from soil samples collected at different locations in Sorocaba/SP (Brazil) and Campinas/SP (Brazil), using the collection strains (obtained from the Phytobacteria Culture Collection of Instituto Biológico (IBSBF, Campinas/SP, Brazil)) *Pseudomonas coronafaciens* pv. *garcae* IBSBF-1372 (PCG-05T), IBSBF-3046 (PCG-06T and PCG-07T) and IBSBF-3270 (PCG-09T) as hosts and the enrichment method described in detail elsewhere (Harada et al., 2022; Kharina et al., 2015; Moreli et al., 2026; Mota et al., 2025; Silva et al., 2023; Zaika et al., 2013), with modifications, and thoroughly characterized from physicochemical, biological and kinetic points of view in a previous work (Moreli et al., 2026).

2.2. Phage virion whole genome sequencing, assembly, taxonomic evaluation, annotation, and phylogeny

The genomes of the 4 isolated bacteriophages were isolated, purified, sequenced, assembled and annotated in a previous work (Moreli et al., 2026), and their sequences are deposited in GenBank NCBI (National Center for Biotechnology Information) under accession numbers PX379591 (phage PCG-05T), PX379592 (phage PCG-06T), PX379593 (phage PCG-07T) and PX379594 (phage PCG-09T). Analyses of the phages' lifestyle were performed using the well-established computational tool BACPHLIP (Hockenberry and Wilke, 2021). a The ViPTree web server was used to generate a "proteomic tree" of viral genome sequences based on genome-wide sequence similarities computed by tBLASTx (Nishimura et al., 2017; Silva et al., 2024; Xuan et al., 2023). To fully clarify the taxonomic and ecological relationships among the four phages, a comprehensive pairwise Average Nucleotide Identity (ANI) (using skani (Shaw and Yu, 2023)) and Average Amino Acid Identity (AAI) (using ezaai (Kim et al., 2021)) analyses were performed across all four novel phages.

2.3. Mechanical properties of the PCG-05T, PCG-06T, PCG-07T and PCG-09T phage genomes

2.3.1. DNA shape

DNA is sequence-dependent, and its study is important in genome-wide analysis. Finding out the structural features of DNA can help reveal the preferred conformations that are intrinsic to a given DNA sequence and its dynamics. The DNA structural features (Propeller Twist (ProT), Helical Twist (HelT), Roll, and Minor groove width (MGW)) were computed independently for the 4 phage genomes (PCG-05T, PCG-06T, PCG-07T, and PCG-09T). Values were obtained using a high-throughput prediction methodology called *DNAshape* (https://rohslab.usc.edu/DNAshape/) (Zhou et al., 2013), following the procedures detailed and described elsewhere (Zhou et al., 2013). In a nutshell, each structural feature was predicted using a pentamer model at a single-nucleotide position using the DNA sequence context of the next base pair. The predicted values were used to infer structural and biophysical information about the genomes. Each DNA structural feature was depicted per genome, showing the predicted DNA shape on the y-axis and the position on the x-axis. Regions with significant deviation were highlighted in the plots. These regions were identified using a modified *z*-score with a Median Absolute Deviation (MAD) of more than 1.5 times. A Python (version 3.9.12) custom script for plotting the resulting heatmaps data was then created and run in Jupyter Notebook (version 6.4.8) within Anaconda Navigator (version 2.1.4, Anaconda Inc.).

2.3.2. Correlation of the DNA structural features

After determining the DNA shape for each genome, a Spearman's rank correlation coefficient analysis, using the R package psych (v2.5.3), was conducted among the DNA shape features within each phage genome. Bivariate scatter plots of matrices with the corresponding features are shown in the y-axis as well in the x-axis. The resulting correlation is shown as a numerical value, with positive values meaning a positive correlation and vice versa. The stars show the *p*-value significance (*p*) calculated from a test of the null hypothesis that the correlation is zero ($H_0 = p = 0$), where *** < 0.001.

2.3.3. Pairwise dinucleotide distance correlation patterns

Sequences from the four phage genomes (PCG-05T, PCG-06T, PCG-07T and PCG-09T) were analyzed to study the dinucleotide distance and correlation patterns. We computed the pairwise distance distribution function following the procedures described elsewhere (Basu et al., 2021b, 2022). The pairwise distance distribution function is a measure of the frequency of two specific dinucleotides that occur at a given separation within the DNA sequence, taking into consideration the spatial arrangement of the dinucleotides. This separation is quantified in terms of nucleotide intervals and normalized using a random expected sequence. We have explored the pairwise distance distribution function of the 256 possible dinucleotide step combinations of AA, AC, AG, AT, CA, CC, CG, CT, GA, GC, GG, GT, TA, TC, TG, and TT. All four phage genomes were analyzed independently but plotted together to allow comparison between the genomes. The resulting plots show the correlation of each dinucleotide step per genome within 100 steps of distance.

2.3.4. Dinucleotide frequency in the PCG-05T, PCG-06T, PCG-07T and PCG-09T phage genomes

A custom R script was used to find the occurrence of the 16 possible dinucleotide combinations (AA, AC, AG, AT, CA, CC, CG, CT, GA, GC, GG, GT, TA, TC, TG, and TT) as a position-independent frequency state in the genome of phages PCG-05T, PCG-06T, PCG-07T and PCG-09T independently. The occurrence of dinucleotides was defined as the number of times a dinucleotide *n* appears in the genome, divided by the length of the genome, and multiplied by percentage. These values were calculated for all phage genomes and plotted as bar plots.

2.3.5. Differential dinucleotide frequency among the genomes

We used a Python (version 3.9.12) custom script to find the pairwise differential dinucleotide frequency to characterize sequence composition differences across phage genomes. Initially, the frequency of occurrence of the dinucleotides in each genome was computed, as detailed before. Subsequently, we constructed a pairwise differential dinucleotide matrix representing the difference in normalized frequency between dinucleotides. These values were visually depicted in a heatmap showing positive and negative values associated with the colors red and blue, respectively.

2.3.6. Phage genome cyclizability

The looping capacity, also called cyclizability, associated with the phage genomes was assessed following the methodology proposed by Basu and colleagues (Basu et al., 2021a, 2021b, 2022) and Zhang and colleagues (Zhang et al., 2020) and fully described elsewhere (Balcão et al., 2022; Harada et al., 2022; Mota et al., 2025; Silva et al., 2024). Intrinsic cyclizability was calculated using a 50-nucleotide interval with a 7-base pair overlap. Cyclizability values were only calculated every 7th base pair, aiming at checking how the bendability changes around some important locations in the phage genome, and to simply average bendability over the entire phage genomes and compare with different phages published in the literature. Cyclizability was assessed independently for all phage genomes. The resultant data were graphically represented per genome to observe the correlation between their predicted cyclizability and their genomic positions; the local mean estimation was also plotted to help visualize the patterns of cyclizability along each genome's position. Local mean estimation was defined as the average value over a local 200 bp window. Additionally, heatmaps and histograms were generated for a visual representation of the distribution of the cyclizability values within the genomes. A Python (version 3.9.12) custom script for calculating genome cyclizability was created and run in Jupyter Notebook (version 6.4.8) within Anaconda Navigator (version 2.1.4, Anaconda Inc.).

2.4. Statistical and computational analyses

Basic descriptive statistics and one-way ANOVA statistical analysis performed to the whole set of cyclizability data were calculated and tabulated using Microsoft Excel (Microsoft, Redmond WA, USA). For the determination of phage virion morphometrics, virion capsid and tail dimensions from 7 individual phage particles per sample were measured and averaged utilizing ImageJ software (version 1.52a, National Institute of Health, USA). To assess sequence-dependent DNA structural features (Propeller Twist, Helical Twist, Roll, and Minor Groove Width), regions presenting significant structural deviations were identified using a modified *z*-score approach. A Median Absolute Deviation (MAD) of greater than 1.5 times was established as the threshold for significance. Additionally, local mean estimations used to visualize patterns of genome cyclizability were defined as the average value calculated over a localized 200 bp window. These computational procedures, alongside data parsing and heatmap generation, were executed via custom Python scripts (version 3.9.12) running in Jupyter Notebook (version 6.4.8) via Anaconda Navigator (version 2.1.4, Anaconda Inc.). Relationships among the predicted DNA structural features within each phage genome were evaluated using Spearman's rank correlation coefficient analysis. This was conducted using the R statistical environment with the psych package (version 2.5.3). The statistical significance of these relationships was established by testing the null hypothesis that the true correlation equals zero ($H_0 = 0$), with statistical significance strictly defined at a *p*-value threshold of < 0.001. Finally, to compare the distributions of non-parametric continuous variables between the different phage genomes (specifically differences in the predicted DNA shape features (ProT, Roll, and MGW) and median intrinsic cyclizability values) a Mann–Whitney *U* test was performed. Differences were considered statistically significant where indicated in the respective analyses.

3. Results

Four newly isolated polyvalent virulent phages preying on *Pseudomonas coronafaciens* pv. *garcae* (Pcg) cells, strains IBSBF-1372, IBSBF-3046 and IBSBF-3270, had their genomes sequenced, annotated and fully analyzed from both molecular and mechanical points of view. All four phages formed clear plaques of lysis on a lawn of the host (Moreli et al., 2026), a hallmark feature of virulent phages, and were previously evaluated aiming at their potential future utilization in coffee plantations for the control of halo blight induced by Pcg.

Based on the morphological analysis of the four phage virions by TEM, all were identified as belonging to class *Caudoviricetes*. Phages PCG-05T, PCG-07T and PCG-09T display myovirus morphotypes whereas phage PCG-06T display a siphovirus morphotype. While phages PCG-05T, PCG-07T and PCG-09T displayed elongated (prolate) icosahedral capsids and a long contractile tail, phage PCG-06T has a perfect icosahedral head and a very long non-contractile tail (Moreli et al., 2026).

3.1. Genomic characterization

The overall metrics of the phage genome assemblies and annotation, such as total size, GC percentage, mapping rate, CheckV evaluation, among other metrics, were described in a previous work (Moreli et al., 2026). All phage genomes were circularized with overlapping of both end sequences, and the assembled phage genomes are complete. The phage genomes were screened for the presence of integrase genes (Kasurinen et al., 2021) and did not feature any putative lysogenic-related (integrase) sequences; since more than half of the predicted protein-coding genes have unknown function, one cannot conclude definitely that these phages have a strictly lytic lifestyle (Hyman, 2019; Kharina et al., 2015). Circular maps of the annotated genomes of phages PCG-05T, PCG-06T, PCG-07T and PCG-09T are displayed in Fig. 1.

Genomic safety profile and functional annotation. To evaluate the safety and suitability of the isolated phages for future biocontrol applications, their completely assembled genomes were systematically screened for deleterious genetic elements. Genomic analysis confirmed the complete absence of known virulence factors, bacterial toxins, and antimicrobial resistance (AMR) genes across all four phages (PCG-05T, PCG-06T, PCG-07T, and PCG-09T). Furthermore, no lysogeny-associated elements, such as integrases, excisionases, or repressor proteins, were detected. The lack of these integration markers substantiates a strictly lytic lifestyle for these newly isolated viral entities.

To characterize the viral genetic repertoire, the predicted protein-coding sequences (CDSs) for each phage were classified into distinct functional modules, including DNA/RNA and nucleotide metabolism, structural assembly (head, tail, and packaging), host lysis, transcription regulation, and auxiliary metabolic genes (AMGs) (Table 1). Consistent with the discovery of deeply branching, novel viral lineages, a substantial proportion of the predicted genes across all four genomes were classified as "viral dark matter", consisting of hypothetical proteins with currently uncharacterized functions.

Comparative functional profiling revealed distinct evolutionary strategies among the isolates. Phages PCG-05T and PCG-07T exhibited identical functional profiles, encoding a streamlined suite of essential replication machinery, standard structural proteins, and a conserved amidase (endolysin) for host cell wall degradation. While phage PCG-09T shared high structural and metabolic synteny with the PCG-05T/PCG-07T lineage, it uniquely encoded a DNA methyltransferase. The presence of this enzyme suggests a specialized evolutionary adaptation, potentially allowing the phage to evade host restriction-modification systems by methylating its own genomic DNA.

In contrast, phage PCG-06T exhibited a highly divergent functional architecture. It displayed a significantly expanded nucleotide metabolism and DNA repair repertoire compared to the other isolates. Notably, its genome encodes thymidylate and thymidine kinases, ribonucleotide reductases, and various exonucleases (e.g., RecJ-like ssDNA

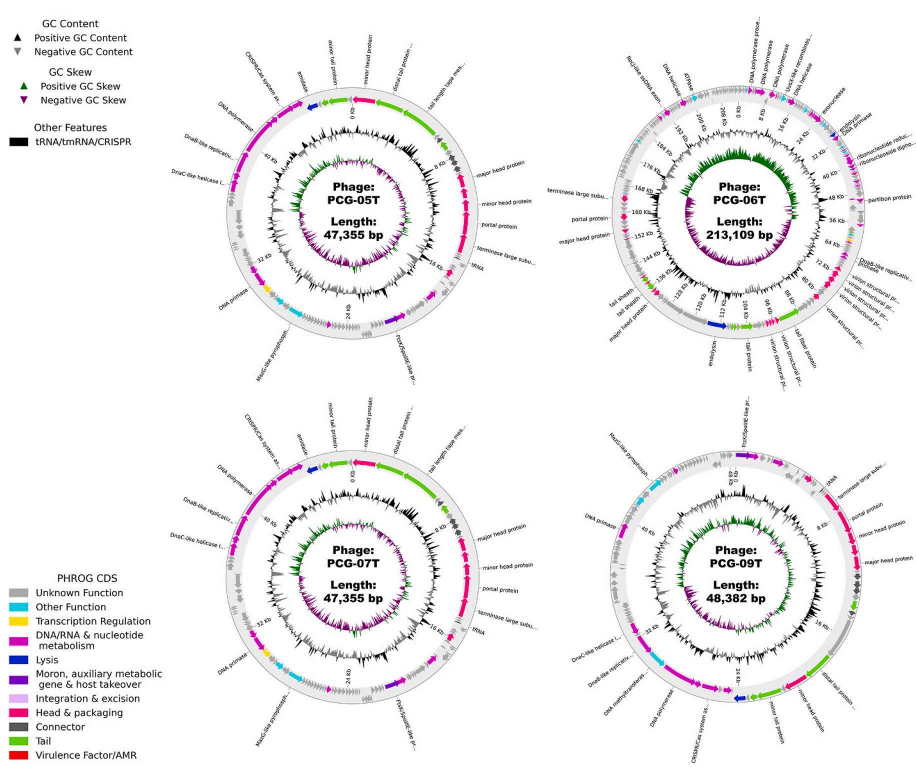

Fig. 1. Annotated genome maps of phages PCG-05T, PCG-06T, PCG-07T and PCG-09T, displaying GC skew, G + C content and predicted CDS. The colored (except light grey) arrows in the outer ring represent the annotated coding sequences (CDSs) according to the annotation performed (Moreli et al., 2026) whereas the light grey arrows correspond to hypothetical proteins, and black arrows correspond to tRNAs. The arrows represent the direction of transcription (strand + or –).

Table 1
Summary of predicted functional gene modules identified in the genomes of phages PCG-05T, PCG-06T, PCG-07T, and PCG-09T.

Functional category	Predicted gene products identified				**Functional description**
	PCG-05T	PCG-06T	PCG-07T	PCG-09T	
Safety profile	No virulence factors, AMR genes, or lysogenic integrases detected				———
Genomic architecture	Displays a highly organized genomic structure typical of strictly virulent phages, equipped with a comprehensive DNA replication and nucleotide metabolism suite	Highly distinct from the other isolates (lineage B). Features a significantly expanded nucleotide metabolism and DNA repair repertoire (including thymidylate/thymidine kinases and an exonuclease), suggesting a high degree of independence from the host's DNA replication machinery	Exhibits 100% amino acid and sequence identity to PCG-05T, representing an independent temporal/ spatial isolation of the exact same viral species. Its functional genetic repertoire is identical to PCG-05T	A close relative to PCG-05T/ PCG-07T (sharing ~86% AAI). While structurally and metabolically highly similar, it distinctively encodes a DNA methyltransferase not found in its sister lineages, potentially indicating an evolved mechanism for restriction-modification evasion within the bacterial host	———
DNA, RNA and nucleotide metabolism	HNH endonuclease, replication initiation protein, thioredoxin domain, DNA primase, DnaC-like helicase loader, DnaB-like replicative helicase, DNA polymerase, CRISPR/Cas system associated, RuvC-like Holliday junction resolvase	DNA polymerase processivity factor, HNH endonuclease, DNA polymerase, DNA binding protein, DNA helicase, exonuclease, DNA primase, replication initiation protein, Holliday junction resolvase, ribonucleotide reductase large subunit, ribonucleoside diphosphate reductase small subunit, dCMP deaminase, partition protein, replication initiation by nicking, RuvC-like Holliday junction resolvase, DnaB-like replicative helicase, primase, nucleotidyltransferase, RecJ-like ssDNA exonuclease	HNH endonuclease, replication initiation protein, thioredoxin domain, DNA primase, DnaC-like helicase loader, DnaB-like replicative helicase, DNA polymerase, CRISPR/Cas system associated, RuvC-like Holliday junction resolvase	Replication initiation protein, HNH endonuclease, RuvC-like Holliday junction resolvase, CRISPR/Cas system associated, DNA polymerase, DnaB-like replicative helicase, DnaC-like helicase loader, DNA primase, thioredoxin domain	Essential enzymes responsible for the replication, repair, and modification of the phage genome during the intracellular infection cycle
Head and packaging	Minor head protein, major head protein, head scaffolding protein, portal protein, terminase large subunit, terminase small subunit	Major head protein, virion structural protein, portal protein, terminase large subunit	Minor head protein, major head protein, head scaffolding protein, portal protein, terminase large subunit, terminase small subunit	Terminase small subunit, terminase large subunit, portal protein, minor head protein, head scaffolding protein, major head protein	Structural proteins that form the viral capsid (head) and the motor complexes required to actively package the dsDNA into the empty procapsid
Tail assembly	Distal tail protein Dit, tail length tape measure protein, major tail protein, tail fiber protein, minor tail protein	Central tail fiber J, tail fiber protein, tail protein, baseplate wedge subunit, tail sheath	Distal tail protein Dit, tail length tape measure protein, major tail protein, tail fiber protein, minor tail protein	Major tail protein, distal tail protein Dit, minor tail protein, tail fiber protein	Proteins forming the tail apparatus, responsible for determining tail length, host recognition (fibers), and initial attachment to the bacterial surface
Connector	Head-tail adaptor, tail completion or Neck1 protein	———	Head-tail adaptor, tail completion or Neck1 protein	Head-tail adaptor, tail completion or Neck1 protein	Structural linkage proteins that seamlessly connect the phage head to the tail apparatus
Lysis	Amidase (endolysin)	Endolysin	Amidase (endolysin)	Amidase (endolysin)	Endolysins responsible for degrading the bacterial peptidoglycan cell wall, allowing the release of newly formed virions
Host takeover and auxiliary metabolic gene (AMGs)	FtsK/SpoIIIE-like protein	Flavodoxin	FtsK/SpoIIIE-like protein	FtsK/SpoIIIE-like protein	AMGs that manipulate host cell metabolism or DNA segregation to favor phage replication
Transcription regulation	RNA polymerase sigma factor			———	Regulatory factors that hijack the host's transcription machinery to control the timing of early/late phage gene expression
Other functions	MazG-like pyrophosphatase, deoxynucleoside monophosphate kinase	Co-chaperonin GroES, UvsX-like recombinase, MazG-like pyrophosphatase, NinI-like serine-threonine phosphatase, ATPase, hydrolase, thymidylate kinase, thymidine kinase, GTP-binding domain	MazG-like pyrophosphatase, deoxynucleoside monophosphate kinase	DNA methyltransferase, deoxynucleoside monophosphate kinase, MazG-like pyrophosphatase	———
Unknown function	Hypothetical proteins				"Viral dark matter" encompassing highly

(continued on next page)

Table 1 (*continued*)

Functional category	Predicted gene products identified				**Functional description**
	PCG-05T	PCG-06T	PCG-07T	PCG-09T	
					divergent or entirely novel proteins with currently uncharacterized functions

exonuclease). The presence of this extensive auxiliary metabolic machinery indicates that phage PCG-06T possesses a high degree of metabolic autonomy, relying less on the host bacterial DNA replication machinery during its infection cycle.

Additionally, to strengthen the analysis of the phages lifestyle, the well-established computational tool BACPHLIP (Hockenberry and Wilke, 2021) was used to provide a more robust phage lifestyle prediction. The results of the lytic probabilities estimated by BACPHLIP were as follows: PCG-05T, Virulent = 0.669, Temperate = 0.331; PCG-06T, Virulent = 0.912, Temperate = 0.088; PCG-07T,

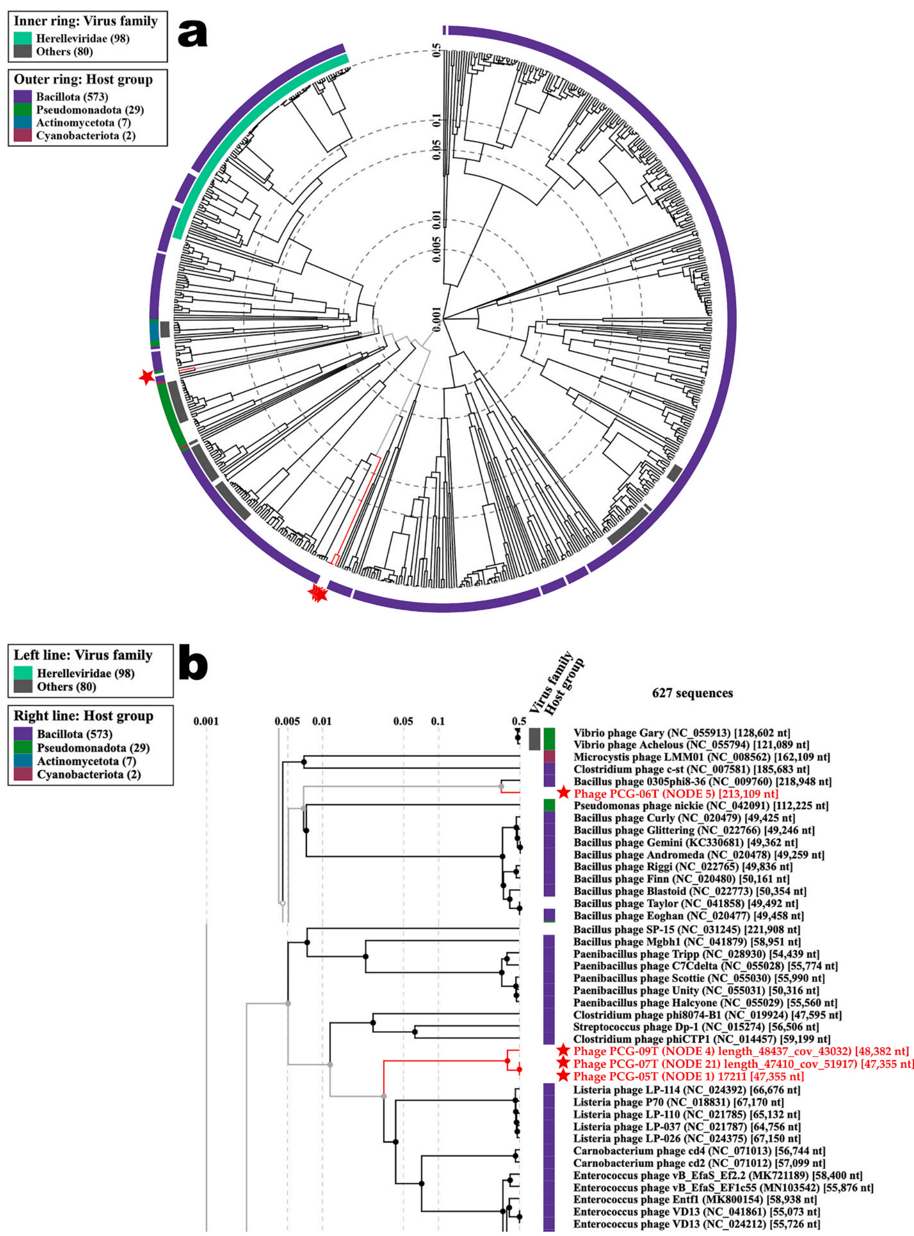

Fig. 2. Viral proteomic tree from ViPTree analysis of phages PCG-05T, PCG-06T, PCG-07T and PCG-09T and related phages. **(a)** 627 phage genomes were used as reference sequences to build phylogenetic trees using ViPTree. In this Figure, phages are identified according to their official ICTV classification, with the inner and outer rings indicating their virus family and host group, respectively. **(b)** Expanded view of the region of the tree containing the most closely related phages. The branches containing phages PCG-05T, PCG-06T, PCG-07T and PCG-09T are displayed in red. The red stars pinpoint the location of phages PCG-05T, PCG-06T, PCG-07T and PCG-09T.

Virulent = 0.669, Temperate = 0.331; PCG-09T, Virulent = 0.812, Temperate = 0.188. Hence, there is strong evidence for all four phages following a virulent lifestyle, with a stronger evidence for phages PCG-06T and PCG-09T.

Using the ViPTree web server (Nishimura et al., 2017), relationships and similarities between phages PCG-05T, PCG-06T, PCG-07T and PCG-09T and other prokaryotic dsDNA viruses were analyzed as described by Silva and colleagues (Silva et al., 2024) and Xuan and colleagues (Xuan et al., 2023). Fig. 2 displays the viral proteomic tree that resulted from ViPTree analysis of phages PCG-05T, PCG-06T, PCG-07T and PCG-09T and related phages.

Results showed that phage PCG-06T was grouped into one small group with another phage (Fig. 2b), viz. Bacillus phage 0305phi8-36, according to the updated ICTV taxonomic classification. Here we can see that phage PCG-06T and phages PCG-05T, PCG-07T and PCG-09T are alone on their branches, showing less phylogenetic similarity (Fig. 2b).

All phages were classified as belonging to the phylum *Uroviricota*, class *Caudoviricetes*. However, the amino acid identity is low with the closest published genomes, and even these genomes (Bacillus phage vB_BthS_BMBphi (Yuan et al., 2018), Bacillus phage BC-7, Bacillus phage BC-6) are classified only up to the class level.

On the network analysis, phages PCG-05T, PCG-07T and PCG-09T were grouped together with Listeria, Carnobacterium and Enterococcus phages (Fig. 2b). Here we can see that phage PCG-06T and phage Bacillus phage 0305phi8-36 (the phage closest to phage PCG-06T) are alone on their branch, showing high phylogenetic similarity (Fig. 2b). According to the analyzes carried out using the ViPTree web server, tBLASTx genomic sequence comparison of phages PCG-05T, PCG-06T, PCG-07T and PCG-09T between themselves and with other phage genomes deposited in the Virus-Host DB (https://www.genome.jp/virushostdb) (Mihara et al., 2016) showed a very high similarity between phages PCG-05T and PCG-07T (Fig. 3a), and between phage PCG-06T and Bacillus phage 0305phi8-36 (NC_009760) (Fig. 3b).

Phage PCG-05T and phage PCG-07T share near-absolute sequence identity and clearly belong to the same viral species based on the established International Committee on Taxonomy of Viruses (ICTV) boundary (≥95% ANI). To fully clarify the taxonomic and ecological relationships among the four phages, a comprehensive pairwise Average Nucleotide Identity (ANI) (using skani (Shaw and Yu, 2023)) and Average Amino Acid Identity (AAI) (using ezaai (Kim et al., 2021)) analyses were performed across all four novel phages. The results are summarized in Table 2.

Taxonomic status (same species, distinct isolates): The quantitative genomic comparison displayed in Table 2 confirms that phages PCG-05T and PCG-07T share 100.00% ANI and 100.00% AAI (sharing all 80 orthologous protein-coding sequences identically). Therefore, from a strict taxonomic standpoint, they represent the same biological species.

Justification for separation as distinct isolates/strains: We categorize them as distinct isolates (or distinct biological strains) rather than duplicate entries because they were recovered from completely independent sampling events or environments and using different bacterial host strains. Phage PCG-05T was isolated from a soil sample collected at the coordinates 23°29′48″S 47°23′20″W in Brazil, using as isolation host the bacterium *Pseudomonas coronafaciens* pv. *garcae* IBSBF-1372 (Moreli et al., 2026). Phage PCG-07T was isolated from a soil sample collected at the coordinates 23°30′00″S 47°24′02″W in Brazil, using as isolation host the bacterium *Pseudomonas coronafaciens* pv. *garcae* Pcg IBSBF-3046 (Moreli et al., 2026). In phage ecology, it is a

Table 2
Pairwise Average Nucleotide Identity (ANI) and Average Amino Acid Identity (AAI) among the four phages. Values above the diagonal represent pairwise AAI (%), and values below the diagonal represent pairwise ANI (%). Self-comparisons are 100.00%.

Phage	PCG-05T	PCG-06T	PCG-07T	PCG-09T
PCG-05T	—	0.00%	**100.00%**	86.18%
PCG-06T	0.00%	—	0.00%	0.00%
PCG-07T	**100.00%**	0.00%	—	86.18%
PCG-09T	90.73%	0.00%	90.73%	—

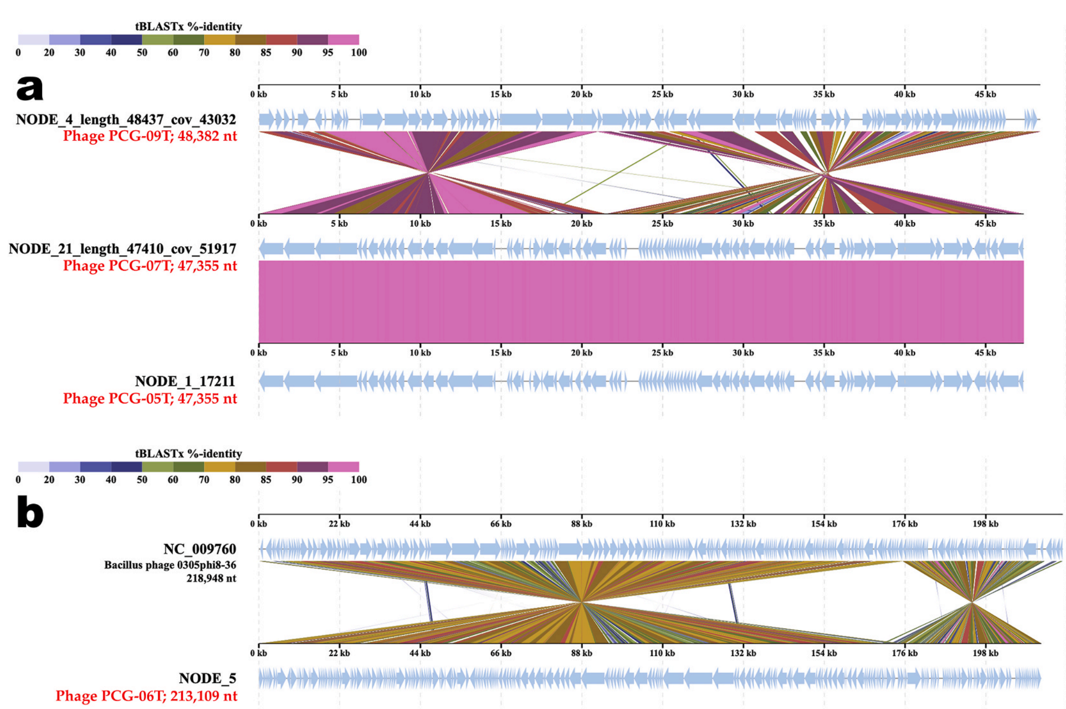

Fig. 3. Genome sequence alignment and comparison of phages PCG-05T, PCG-07T and PCG-09T **(a)** and of phage PCG-06T with another phage genome exhibiting co-linearity **(b)** detected by TBLASTX using the ViPTree web server (Nishimura et al., 2017). Homologous regions detected by a tBLASTx search are connected by segments colored based on amino acid identity. The upper color bar shows the % identity of tBLASTx.

well-documented phenomenon to isolate identical or near-identical viral genotypes from spatially or temporally separate environments due to stable environmental persistence or clonal expansion.

Broader genomic context: The matrix in Table 2 also reveals that phage PCG-09T is a closely related sister species sharing 90.73% ANI and 86.18% AAI with the PCG-05T/PCG-07T lineage, while phage PCG-06T shares 0.00% identity with the rest, highlighting its status as an entirely unique and unrelated phage group.

3.2. DNA structural features of phages PCG-05T, PCG-06T, PCG-07T and PCG-09T

In a previous work, the four lytic polyvalent phages displayed very different characteristics in the adsorption, infection and virion morphogenesis processes (Moreli et al., 2026) within their isolation host strains (*Pseudomonas coronafaciens* pv. *garcae*, Pcg) (PCG-05T and Pcg IBSBF-1372, PCG-06T and Pcg IBSBF-3046, PCG-07T and Pcg IBSBF-3046, and PCG-09T and Pcg IBSBF-3270). Hence, studying the mechanical properties of their genomes (three myoviruses and one siphovirus) was essential for a better and deeper understanding of the enormous variability in the infection rates of their target bacterial cells with concomitant very discrepant yields of the virion morphogenesis process within infected cells (viz. 93 virions per host cell (phage PCG-05T) *versus* 10 virions per host cell (phage PCG-06T) *versus* 80 virions per host cell (phage PCG-07T) *versus* 104 virions per host cell (phage PCG-09T), results from Moreli and colleagues (Moreli et al., 2026)). Fig. 4 displays the statistical characteristics of the 4 predicted structural features from the PCG-05T, PCG-06T, PCG-07T and PCG-09T phage genomes.

Table 3 displays the PCG-05T, PCG-06T, PCG-07T and PCG-09T phage dsDNA sequence statistics.

Fig. 5 displays the results from DNA shape (Propeller Twist, Minor Groove Width, Roll and Helical Twist) calculations for the assembled genomes of phages PCG-05T, PCG-06T, PCG-07T and PCG-09T.

In the present study, the DNA shape features of phages PCG-05T, PCG-06T, PCG-07T and PCG-09T genomes were predicted using a high-throughput method called DNAShape (a web-based tool), with the outcomes being visually represented as heatmaps (Fig. 5). ProT refers to the angle between the planes of two DNA bases. When the nearer base rotates clockwise, the angle is considered positive; counterclockwise rotation yields a negative angle. This measurement reflects DNA's flexibility, with a larger ProT indicating a more rigid DNA structure (El Hassan and Calladine, 1996). The presence of propeller twists in base pairs complicates their stacking into dinucleotide steps compared to in-plane base pairs. Specifically, it creates a chemical locking effect, increasing the rigidity of dinucleotide steps with high ProT angles. Therefore, smaller ProT facilitate easier sliding of base pairs over each other within the constraints of the DNA backbone (El Hassan and Calladine, 1996). In the ProT heatmaps (Fig. 5), it is observed higher and lower ProT regions within the phage genome sequences. The higher and lower values are represented by yellow and dark blue colors, respectively (Fig. 5, ProT). Notably, higher average ProT angles are observed in the PCG-05T phage genome (Fig. 5, ProT, with a mean of −7.421°), PCG-06T phage genome (Fig. 5, ProT, with a mean of −7.580°), PCG-07T phage genome (Fig. 5, ProT, with a mean of −7.422°), and PCG-09T phage genome (Fig. 5, ProT, with a mean of −7.404°), in contrast to the values reported by Silva and colleagues (Silva et al., 2024) for two myoviruses also lytic for Pcg (−8.21 for phage PsgM02F and −7.56 for phage PsgM04F), suggesting that on average there are more flexible regions in PCG-05T, PCG-06T, PCG-07T and PCG-09T phage genomes than in those two myoviruses genomes. However, while ProT values may be comparable on average across those phage genomes, variations can arise at the local level due to differences in the distribution of ProT within each genome.

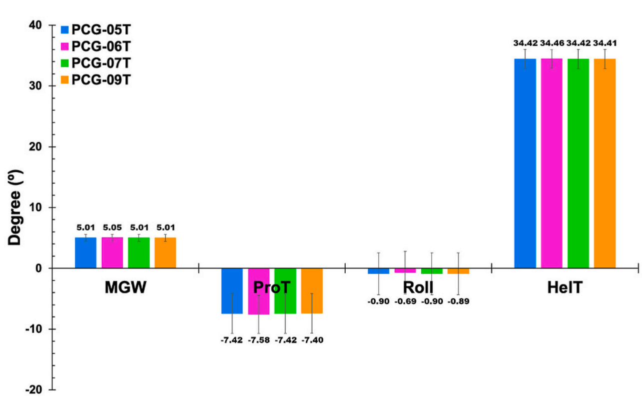

Fig. 4. Statistical characteristics of the 4 predicted structural features from the PCG-05T, PCG-06T, PCG-07T and PCG-09T phage genomes.

The DNA helical structure arises from the stacking of individual bases on top of each other, with the bases twisting slightly to optimize their hydrophobic interactions, thereby forming the helix (Fratini et al., 1982). The flexibility of DNA twist plays a crucial role in its various biological functions. Previous studies have examined this flexibility using short-base sequences due to their looping ability (Cohen and Meselson, 1988; Zhou et al., 2015), as well as investigating the bending and twisting deformations necessary for protein-DNA interactions (Crothers et al., 1992; Shore et al., 1981; Zhang and Crothers, 2003). Collectively, these findings suggest that HelT serves as an effective metric for DNA flexibility. The evaluation of HelT heatmap in PCG-05T, PCG-06T, PCG-07T and PCG-09T phage genomes revealed high HelT degree (Fig. 5, HelT), with mean values of 34.420° (PCG-05T), 34.456° (PCG-06T), 34.420° (PCG-07T) and 34.411° (PCG-09T), comparable to the values reported by Silva and colleagues (Silva et al., 2024) for two myoviruses (−34.61 for phage PsgM02F and 34.49 for phage PsgM04F). The HelT heatmaps provide insights into the distribution of HelT angles in the phage genomes, with dark blue indicating a higher angle and white representing a lower HelT angle (Fig. 5, HelT).

The roll angle serves as a local angular parameter delineating the bending and twisting of adjacent base pairs (Richmond and Davey, 2003; Zhurkin et al., 1979). Notably, significant positive roll values in base pair steps indicate weak stacking interactions (Zhou et al., 2015). Tolstorukov and colleagues (Tolstorukov et al., 2007) proposed a correlation between roll and conformational flexibility in the nucleosome. Through molecular dynamic simulations, those researchers demonstrated that setting the roll value to zero at all base-pair steps nearly straightens the DNA molecule, illustrating that those variations in roll accounted for DNA bending. Furthermore, they observed that negative roll values cause the DNA to bend towards the minor groove width, while positive roll values induce bending towards the major groove width. These findings underscore the relationship between roll and DNA bendability.

The Roll angle values of the PCG-05T, PCG-06T, PCG-07T and PCG-09T phage genomes are displayed in the heatmaps (Fig. 5, Roll) with a yellow color indicating a higher degree, and a dark brown color indicating a lower degree of Roll (Fig. 5, Roll). Indeed, we can discern variations in the distribution of roll angles in the PCG-05T, PCG-06T, PCG-07T and PCG-09T phage genomes. The average Roll value is −0.899° (PCG-05T), −0.693° (PCG-06T), −0.900 (PCG-07T) and −0.894 (PCG-09T), comparable to the values reported by Silva and colleagues (Silva et al., 2024) for their two myoviruses (−0.89 for phage PsgM02F and −0.82 for phage PsgM04F). The roll angles for these phage genomes are negative, with a minor difference. A less negative roll angle value for the PCG-06T genome might indicate a lower degree of bending towards the minor groove width in certain regions of the phage genome when compared with the PCG-05T, PCG-07T and PCG-09T phage genomes, but more studies are needed to prove this statement.

The minor groove width (MGW) varies geometrically with sequence and can significantly influence DNA shape recognition by proteins

Table 3
PCG-05T, PCG-06T, PCG-07T and PCG-09T phage dsDNA sequence statistics.

Phage	Genome length (bp)	A (bp)	T (bp)	G (bp)	C (bp)	AT (bp)	GC (bp)	GC content (%)	AT content (%)
PCG-05T	47355	**12975**	15132	9106	**10142**	**28107**	**19248**	**40.65**	**59.35**
PCG-06T	213109	61499	62520	44908	44182	124019	89090	41.80	58.20
PCG-07T	47355	**12976**	15132	9106	**10141**	**28108**	**19247**	**40.64**	**59.36**
PCG-09T	48382	15249	13481	10236	9416	28730	19652	40.62	59.38

Notes: adenine (A), thymine (T), guanine (G), cytosine (C); base pairs (bp).

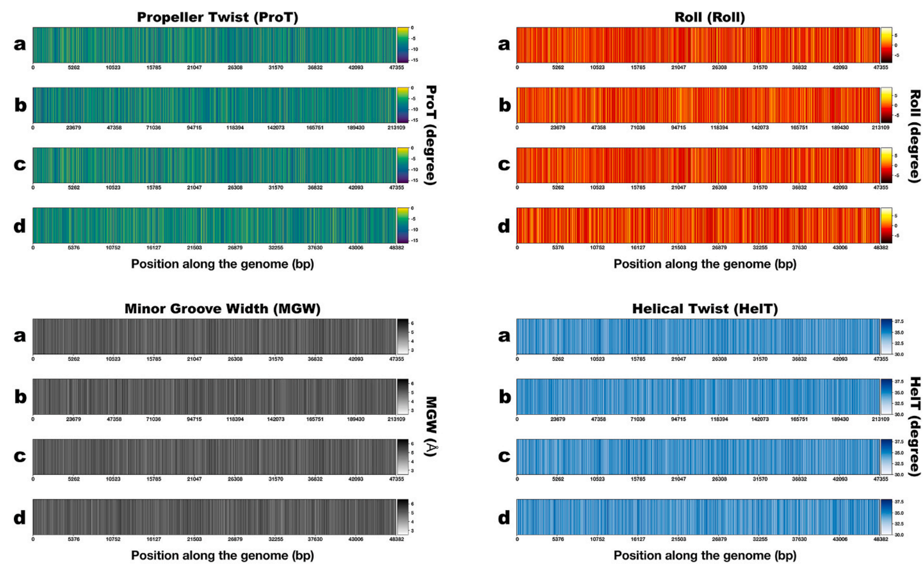

Fig. 5. Heatmap patterns from the results of DNA shape (Propeller Twist, Minor Groove Width, Roll and Helical Twist) calculations for the assembled genomes of phages PCG-05T **(a)**, PCG-06T **(b)**, PCG-07T **(c)** and PCG-09T **(d)**.

(Wing et al., 1980). A narrower minor groove suggests greater accessibility of base edges to proteins, such as transcription factors (Zhou et al., 2015). This might be due to the groove width compression caused by the unusually close spacing between recognition helices of each helix-turn-helix motif (Kostrewa et al., 1991). On the other hand, a study conducted by Oguey and colleagues (Oguey et al., 2010) demonstrated a relationship between flexibility and MGW, suggesting that this relationship is likely an intrinsic property of DNA. The MGW can be also visualized in the respective heatmap (Fig. 5, MGW), where black indicates a higher MGW angle value, while white indicates a lower MGW angle value. The distribution of black and white patterns varies along the phage genome sequences, highlighting the varying significance of MGW in different regions of the phage genomes at the local level. The mean MGW angle value was 5.011° (PCG-05T), 5.048° (PCG-06T), 5.011° (PCG-07T) and 5.014° (PCG-09T), comparable to the value reported by Mota and colleagues (Mota et al., 2025) for a siphovirus (5.040° for phage PcgS01F) and comparable to the values reported by Silva and colleagues (Silva et al., 2024) for two myoviruses (4.98 for phage PsgM02F and 5.02 for phage PsgM04F). The PCG-06T phage genome exhibits a wider and less concentrated MGW value compared to those values of phages PCG-05T, PCG-07T, PCG-09T, to the value of phage PcgS01F reported by Mota and colleagues (Mota et al., 2025) and to those values of phages PsgM02F and PsgM04F reported by Silva and colleagues (Silva et al., 2024). The less heightened narrowness of the MGW in the PCG-06T phage genome may suggest decreased protein-DNA interactions and less bending in certain regions of the genome, as proteins like transcription factors tend to induce DNA bending, as opposed to phages PCG-05T, PCG-07T, PCG-09T, PsgS01F (Mota et al., 2025) and PsgM02F (Silva et al., 2024). A discernible difference in the MGW between the PCG-05T, PCG-06T, PCG-07T and PCG-09T phage genomes can be noticed, which might indicate a difference in MGW in the genomes at nucleotide and global level, with a narrower MGW for the PCG-05T and PCG-07T genomes.

After analyzing the DNA shape of the PCG-05T, PCG-06T, PCG-07T, and PCG-09T phage genomes, a Spearman correlation analysis was conducted on their DNA mechanical features, to explore the linear relationship among the DNA mechanical features. The results were visually represented in four graphs, displaying the Spearman correlation values for each structural feature (Fig. 6).

The Spearman correlation analysis between the four predicted structural features of the four phage genomes revealed both negative and positive correlations. Notably, positive correlations were observed between the Roll angle degree and the MGW (Fig. 6), with a value of 0.54 (PCG-05T genome), 0.51 (PCG-06T genome), 0.54 (PCG-07T genome) and 0.55 (PCG-09T genome). This relationship aligns with previous research by Fratini and colleagues (Fratini et al., 1982), which demonstrated that the roll angle measures the degree to which successive base pairs open towards the minor groove. Consequently, higher positive roll angles may yield wider minor groove widths. Conversely, a negative correlation was observed between the HelT and the MGW (Fig. 6), with a value of −0.24° (PCG-05T genome), −0.23° (PCG-06T genome), −0.24° (PCG-07T genome), and −0.22° (PCG-09T genome). Liebl and colleagues (Liebl et al., 2015) demonstrated that DNA overwinding decreases the MGW, leading to an increased HelT due to enhanced stacking distances between adjacent base pairs. Thus, we can infer a probable negative correlation between HelT and MGW.

Moreover, we observed positive and negative correlations among DNA structural features, such as the positive correlation between Roll and HelT, which measured 0.02 (PCG-05T genome), 0.04 (PCG-06T genome), 0.02 (PCG-07T genome), and 0.02 (PCG-09T genome) (Fig. 6). Although the Spearman correlation, in this case, is close to zero, indicating a minimal association between these features, prior research has identified an anti-correlation likely due to the presence of specific pyrimidine/purine steps in the genome (Kannan et al., 2006; Olson et al.,

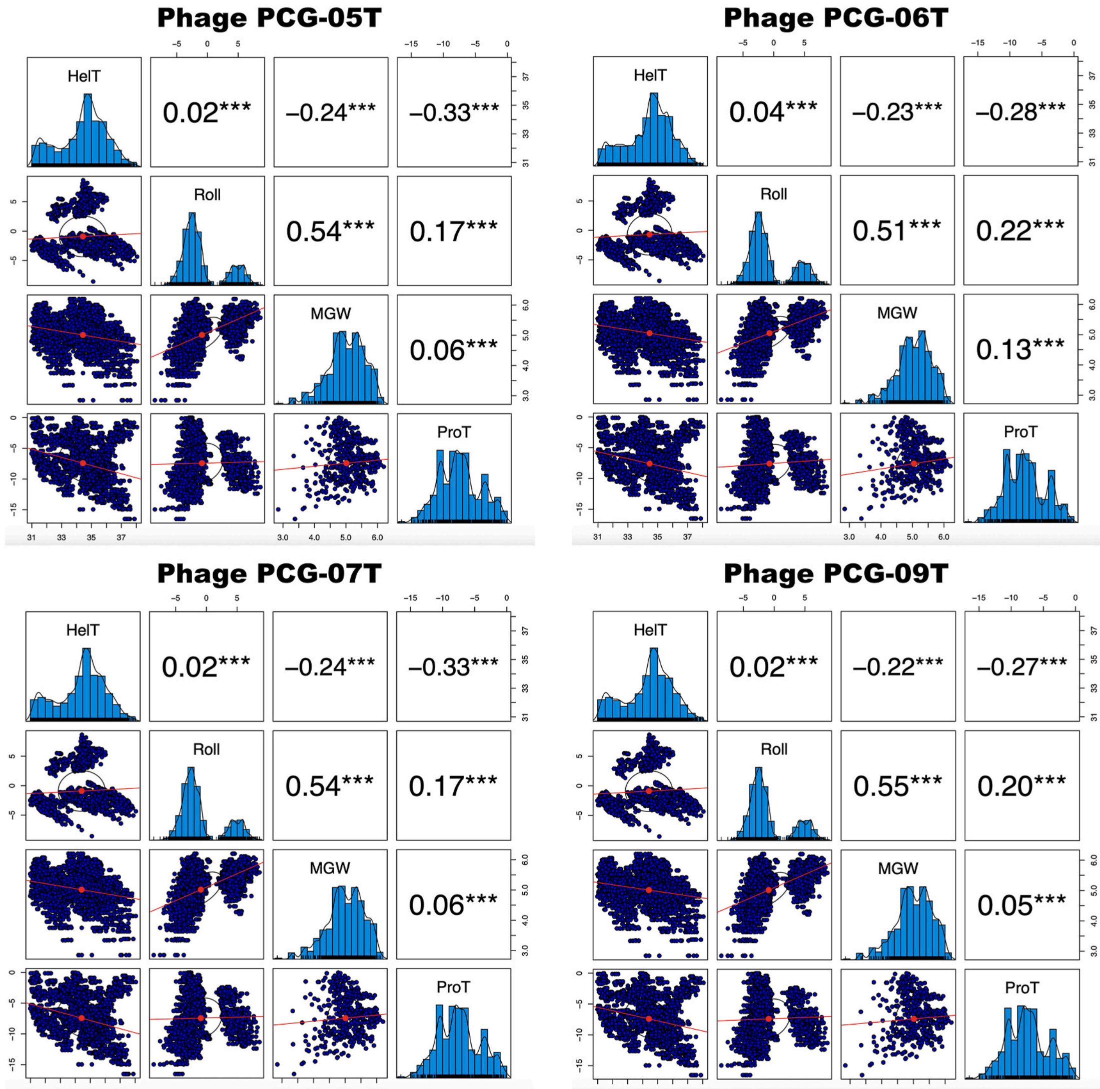

Fig. 6. Spearman correlations between the 4 predicted structural features of the genomes of phages PCG-05T, PCG-06T, PCG-07T, and PCG-09T.

1998; Packer et al., 2000).

Furthermore, a negative correlation between ProT and HelT was observed in the phage genomes, with a value of −0.33 (PCG-05T genome), −0.28 (PCG-06T genome), −0.33 (PCG-07T genome), and −0.27 (PCG-09T genome) (Fig. 6). Propeller twisting aims to optimize stacking interactions. When ProT values approach 36°, they tend towards zero, indicating optimal stacking interactions. Deviations might result in diminished stacking. El Hassan and Calladine (El Hassan and Calladine, 1996) noted that more negative ProT angles in double-stranded DNA (dsDNA) are associated with high flexibility, whereas ProT angles ranging from −1 to −3° are linked to greater rigidity in dsDNA. This statement is supported by Basu and colleagues (Basu et al., 2022), who also showed that high ProT correlates with low cyclizability, indicative of reduced DNA bendability.

Based on these findings, we can infer that the PCG-05T, PCG-06T, PCG-07T and PCG-09T phage genomes show a degree of correlation among the DNA structural features which might influence the mechanical properties of the DNA. DNA shape could potentially impact the biology of the phages. Narrow MGW, distinctive HelT conformations, alterations in roll, and variations in ProT may collectively affect various biological processes associated with the phages, such as DNA translocation. This could happen by facilitating enhanced DNA-protein interactions, improving accessibility, and contributing to the stability of DNA structure during the translocation event.

While our findings suggest significant differences between the PCG-05T, PCG-06T, PCG-07T and PCG-09T phage genomes, it is important to note that these analyses were primarily designed to study the impact at the local level rather than at the global level. Therefore, further studies are necessary to validate these observations and understand their broader implications.

A Python script was used to investigate the occurrence of the possible 16 dinucleotides (AA, AC, AG, AT, CA, CC, CG, CT, GA, GC, GG, GT, TA, TC, TG, and TT) in the genomes of phages PCG-05T, PCG-06T, PCG-07T and PCG-09T. This was accomplished by counting the occurrences of each dinucleotide in each phage genome and dividing them by the respective genome's length. Fig. 7 displays the dinucleotide frequency in the genomes of phages PCG-05T, PCG-06T, PCG-07T and PCG-09T.

The frequency of occurrence of dinucleotides (AA, AC, AG, AT, CA, CC, CG, CT, GA, GC, GG, GT, TA, TC, TG, and TT) in the PCG-05T, PCG-06T, PCG-07T and PCG-09T phage genomes was calculated and displayed in a bar graph (Fig. 7), exhibiting the percentage of each dinucleotide in the phage genomes. Notably, certain dinucleotides exhibit higher frequencies in the phage genomes. Dinucleotides AT, CT, and TA demonstrate the highest frequencies in the genomes (Fig. 7). Conversely, the dinucleotides CC, CG, GG, and GC exhibit the lowest frequencies. The CT dinucleotide exhibits the highest occurrence frequency in the

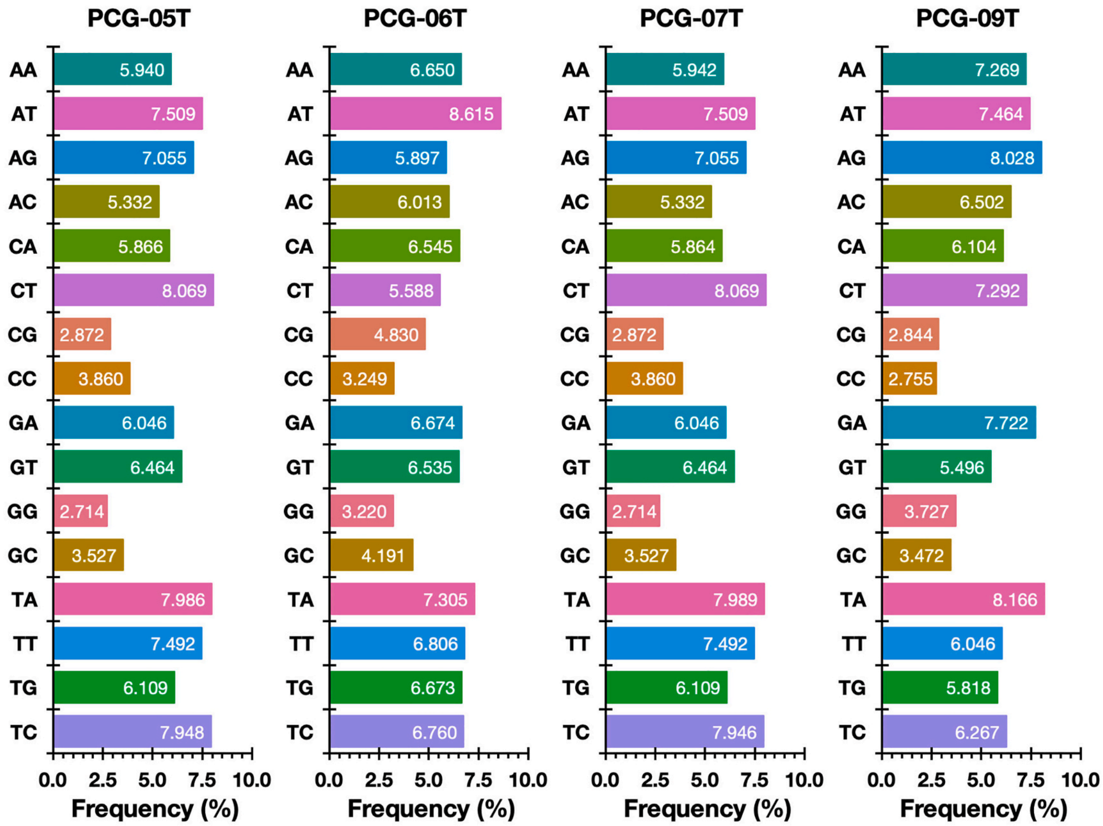

Fig. 7. Dinucleotide frequency in the genomes of phages PCG-05T, PCG-06T, PCG-07T and PCG-09T.

PCG-05T and PCG-07T phage genomes, reaching 8.069%, followed closely by the TA dinucleotide with the second-highest occurrence frequencies in these two phage genomes at 7.986% (PCG-05T) and 7.989% (PCG-07T) (Fig. 7). In contrast, the CG dinucleotide displays the lowest occurrence frequency in both phage genomes, accounting for 2.872%, followed closely by GG dinucleotide (2.714% in both genomes). Phage PCG06T displayed the highest frequency of AT (viz. 8.615%) whereas its TA, CT and AG frequencies were the lowest of the four phage genomes

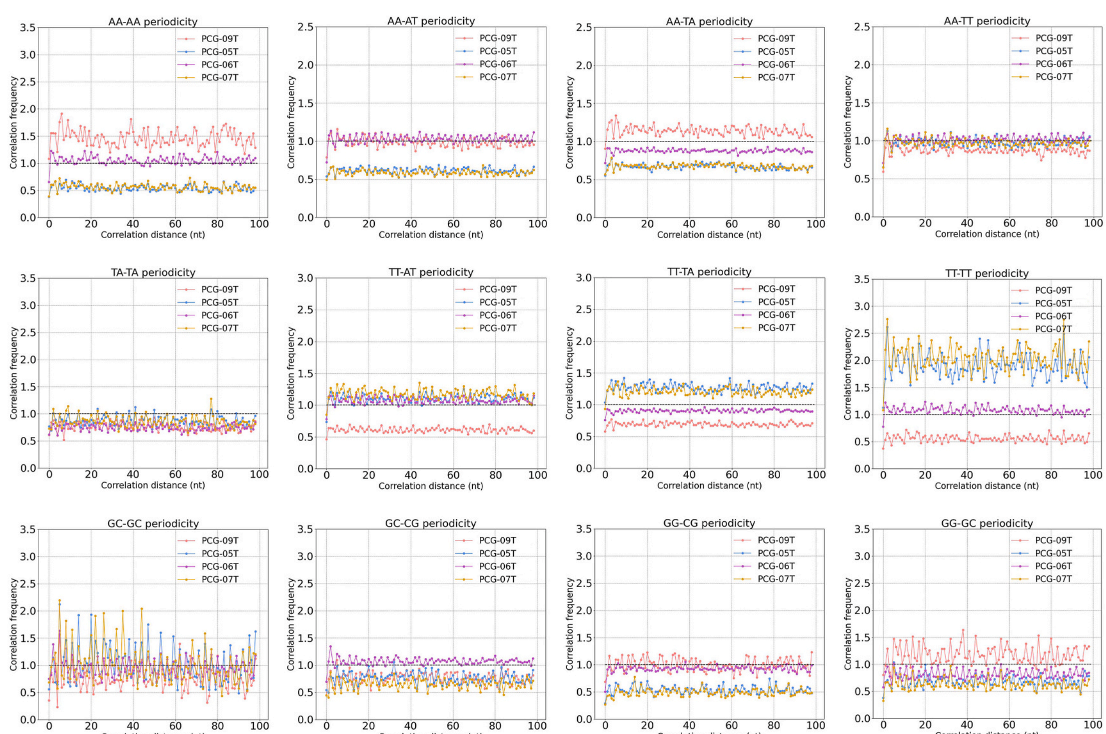

Fig. 8. Cross dinucleotide pairwise distance correlation patterns in the assembled genomes of phages PCG-05T, PCG-06T, PCG-07T and PCG-09T.

(viz. 7.305%, 5.588% and 5.897%, respectively). Phage PCG-09T, on the other hand, displayed the lowest CC, CG, GC, TC, TG, TT, and AT (2.755%, 2.844%, 3.472%, 6.267%, 5.818%, 6.046% and 7.464%, respectively) and the highest AC, AG, and AA (6.502%, 8.028% and 7.269%, respectively).

Fig. 8 displays the results obtained in the most significant cross dinucleotide pairwise distance correlation calculations performed to the four phage genomes. With these calculations, one aimed at checking the presence of peak patterns meaning periodicity. The X-axis in the plots in Fig. 8 represent the number of incremental steps where one can observe such periodicity.

Initially, one determined the frequency of all 16 possible dinucleotide combinations independently in all four phage genomes (Fig. 7). Subsequently, the difference in dinucleotide frequencies between phage genome pairs were computed by counting the occurrence of each dinucleotide in each phage genome and dividing it by the respective genome length, followed by determining the differences in dinucleotide frequencies between the aforementioned phage genome pairs and representing them as heatmaps (Fig. 9), where a high differential frequency can be seen depicted in red, whereas a small differential frequency can be seen depicted in blue.

Fig. 10 displays the heatmap patterns of the predicted intrinsic cyclizability values of the genomes of phages PCG-05T, PCG-06T, PCG-07T and PCG-09T.

In Fig. 11, the predicted intrinsic cyclizability along the phage genome at 7 bp resolution can be observed for phages PCG-05T **(a)**, PCG-06T **(b)**, PCG-07T **(c)** and PCG-09T **(d)**.

Fig. 12 allows to observe a plot with the statistics of the cyclizabilities of the four phage genomes.

In Fig. 10, the heatmap patterns of the predicted intrinsic cyclizability along the phage genome at 7 bp resolution can be observed for phages PCG-05T **(a)**, PCG-06T, PCG-07T and PCG-09T, following the procedures detailed by Basu and colleagues (Basu et al., 2022). Their approach involved the simultaneous measurement of intrinsic cyclizability across a broad spectrum of 90,000 distinct 50 base pair DNA sequences.

Notably, the cyclizability pattern varies along the phage genomes (Fig. 10a–d), revealing distinct peaks of cyclizability (Fig. 11). In the PCG-05T phage genome (Fig. 11), there are some peaks reaching a maximum value of 0.677 and a minimum of −0.766. In the PCG-07T phage genome (Fig. 11), there are some peaks reaching a maximum value of 0.677 and a minimum of −0.766. In the PCG-09T phage genome (Fig. 11), there are some peaks reaching a maximum value of 0.711 and a minimum of −0.672. However, despite the observed higher cyclizability peaks in the PCG-05T, PCG-07T and PCG-09T phage genomes, the mean cyclizability values for these genomes are −0.10803 (PCG-05T), −0.10788 (PCG-07T) and −0.10618 (PCG-09T), values that are higher to those reported by Silva and colleagues (Silva et al., 2024) for two myoviruses (with an average of −0.12567 for the PsgM02F phage genome and −0.16482 for the PsgM04F phage genome). This suggests that, on average, the PCG-05T, PCG-07T and PCG-09T phage genomes have higher cyclizabilities compared to those of the PsgM02F and PsgM04F phage genomes. Overall, these results might suggest that the PCG-05T, PCG-07T and PCG-09T phage genomes are somewhat more flexible than the PsgM02F and PsgM04F phage genomes. Regarding the PCG-06T phage genome (Fig. 11), there are some peaks reaching a maximum value of 0.679 and a minimum of −0.922. However, despite the observed higher cyclizability peaks in the PCG-06T phage genome, the mean cyclizability value for this genome is −0.19807 (PCG-06T), a value that is lower than those reported by Silva and colleagues (Silva et al., 2024) for two myoviruses (with an average of −0.12567 for the PsgM02F phage genome and −0.16482 for the PsgM04F phage genome). This suggests that, on average, the PCG-06T phage genome has a lower cyclizability compared to those of the PCG-05T, PCG-07T and PCG-09T phage genomes (Fig. 12) and to those of PsgM02F and PsgM04F phage genomes, being somewhat more rigid than the PCG-05T, PCG-07T, PCG-09T, PsgM02F and PsgM04F phage genomes.

Fig. 13 allows to observe the distribution of cyclizability values' sizes and frequency of apparition in the PCG-05T, PCG-06T, PCG-07T and PCG-09T phage genomes.

As we can see (Fig. 13), the distribution of cyclizability from the PCG-05T, PCG-06T and PCG-07T genomes goes from ca. −0.80 to 0.75, with median values of −0.1143 (PCG-05T), −0.2055 (PCG-06T) and −0.1140 (PCG-07T), whereas the distribution of cyclizability from the

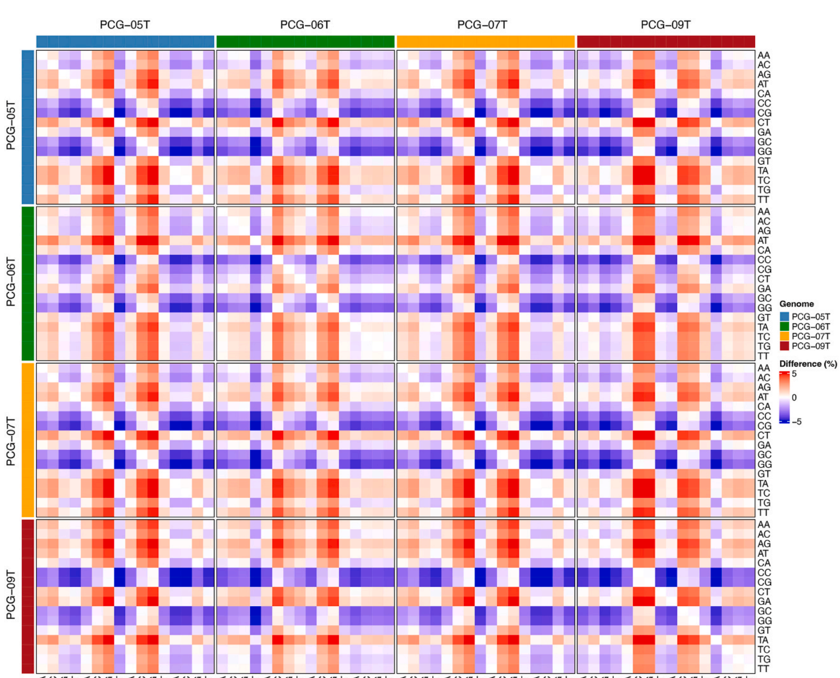

Fig. 9. Heatmaps of the differential dinucleotide frequencies between phage genome pairs PCG-05T vs. PCG-06T, PCG-05T vs. PCG-07T, PCG-05T vs. PCG-09T, PCG-06T vs. PCG-07T, PCG-06T vs. PCG-09T, and PCG-07T vs. PCG-09T.

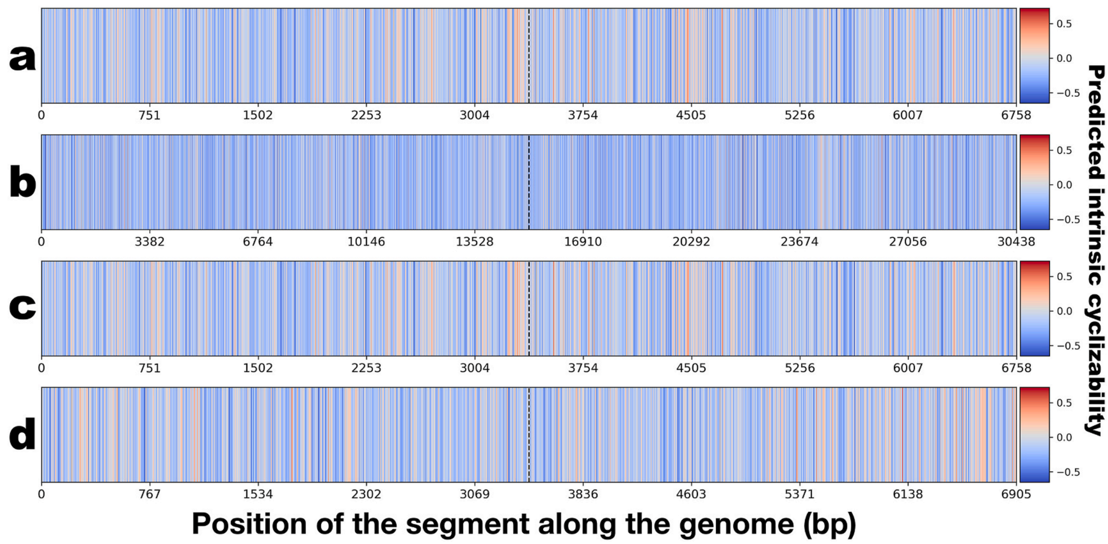

Fig. 10. Heatmap patterns of the cyclizability values of the genomes of phages PCG-05T **(a)**, PCG-06T **(b)**, PCG-07T **(c)** and PCG-09T **(d)**.

PCG-09T genome goes from ca. −0.65 to 0.75, with a median value of −0.1175 (PCG-09T). Based on these distributions we can suggest that there are more peaks with a high cyclizability (close to zero) in the PCG-05T, PCG-07T and PCG-09T phage genomes than in the PCG-06T phage genome.

A close inspection of the data in Fig. 12 allows to observe relatively large standard deviations of the mean predicted cyclizabilities and, to check if the average predicted intrinsic cyclizabilities of phages PCG-05T, PCG-06T, PCG-07T and PCG-09T genomes were statistically different between them, a one-way ANOVA statistical analysis was performed to the whole set of genome predicted intrinsic cyclizability data, to test the null hypothesis that the average predicted intrinsic cyclizabilities of the four phage genomes were similar. Table 4 displays the statistics of the predicted intrinsic cyclizability data of the genomes of phages PCG-05T, PCG-06T, PCG-07T and PCG-09T, whereas the results obtained in the one-way ANOVA statistical analysis performed to the whole set of data are displayed in Table 5.

The *F*-ratio allows to answer the question of whether the variance between the means of the predicted intrinsic cyclizability populations of the four phage genomes were significantly different, whereas the *p*-value is the probability of getting average predicted intrinsic cyclizabilities at least as extreme as the ones that were actually observed in the genomes of the four phages.

Phage PCG-09T displayed the highest average predicted intrinsic cyclizability (−0.106180058) and also produced the largest virion progeny inside infected host cells (104.22 virions per host cell), whereas phage PCG-06T displayed the lowest average predicted intrinsic cyclizability (−0.198067586) and also produced the lowest virion progeny inside infected host cells (9.77 virions per host cell). Phages PCG-07T (average predicted intrinsic cyclizability of −0.107877717) and PCG-05T (average predicted intrinsic cyclizability of −0.108032635) displayed slightly different average predicted intrinsic cyclizabilities, and also slightly different virion morphogenesis yields inside infected host cells, viz. 80.34 virions per host cell (PCG-07T) and 92.69 virions per host cell (PCG-05T). This trend confirms the previous postulated hypothesis that a phage genome with higher predicted intrinsic cyclizability is more flexible (bendable) and therefore, once translocated into the infected host cell cytoplasm, the RNA-polymerase inside the host cell negotiates better with it and leads to more extensive dsDNA transcription and concomitant production of phage proteins in larger amounts leading to higher virion morphogenesis yields.

4. Discussion

Designing effective and eco-conscious alternatives to traditional copper- and antibiotic-based treatments for managing *Pseudomonas coronafaciens* pv. *garcae* (Pcg) infections in coffee crops remains a significant scientific hurdle. In this study, we report the comprehensive analysis of four newly isolated polyvalent lytic bacteriophages (PCG-05T, PCG-06T, PCG-07T, and PCG-09T) capable of targeting Pcg. Each phage was examined not only for its virion production efficiency within infected host cells but also underwent full genomic profiling. This included both molecular annotation and assessment of DNA mechanical features, offering new insights into the biological performance and structural dynamics of these candidate biocontrol agents.

Morphological characterization of phages PCG-05T, PCG-06T, PCG-07T, and PCG-09T by transmission electron microscopy (Moreli et al., 2026) and whole genome sequencing confirmed phages PCG-05T, PCG-07T, and PCG-09T as myoviruses (phage morphotypes with elongated (prolate) icosahedral capsids and a long contractile tail) and phage PCG-06T as a siphovirus (perfect icosahedral head and a very long non-contractile tail). All phages belong to the class *Caudoviricetes*, according to recent bacterial viruses' taxonomy (Turner et al., 2023). The taxonomic lineage of phages PCG-05T, PCG-06T, PCG-07T, and PCG-09T was therefore established as *Viruses* > *Duplodnaviria* > *Heunggongvirae* > *Uroviricota* > *Caudoviricetes* (Schoch et al., 2020; UniProt Consortium, 2021), with family/genus remaining undetermined. The genomes of all phages were screened for the presence of integrase genes following whole phage genome sequencing and annotation (Hyman, 2019) and did not feature any lysogenic-related (integrase) genes, but since more than half of the predicted protein-coding genes have unknown function (Moreli et al., 2026) one cannot conclude definitely that these phages have a strictly lytic lifestyle (Balcão et al., 2022; Hyman, 2019; Kakasis and Panitsa, 2019; Kharina et al., 2015). The genomic characterization of phages PCG-05T, PCG-06T, PCG-07T, and PCG-09T underscores their profound evolutionary novelty and their distinct potential as biocontrol agents. A defining hallmark of these genomes is the substantial presence of “viral dark matter”, with over half of the predicted protein-coding sequences possessing completely unknown functions (Fig. 1). Crucially, the complete absence of integrase genes or putative lysogenic-related sequences within these fully circularized genomes strongly implies a strictly virulent lifestyle, fulfilling a primary prerequisite for the safe application of phages in agricultural or clinical settings. From a taxonomic perspective, these isolates exhibit deep evolutionary divergence. Although classified within the class *Caudoviricetes*, their remarkably low amino acid

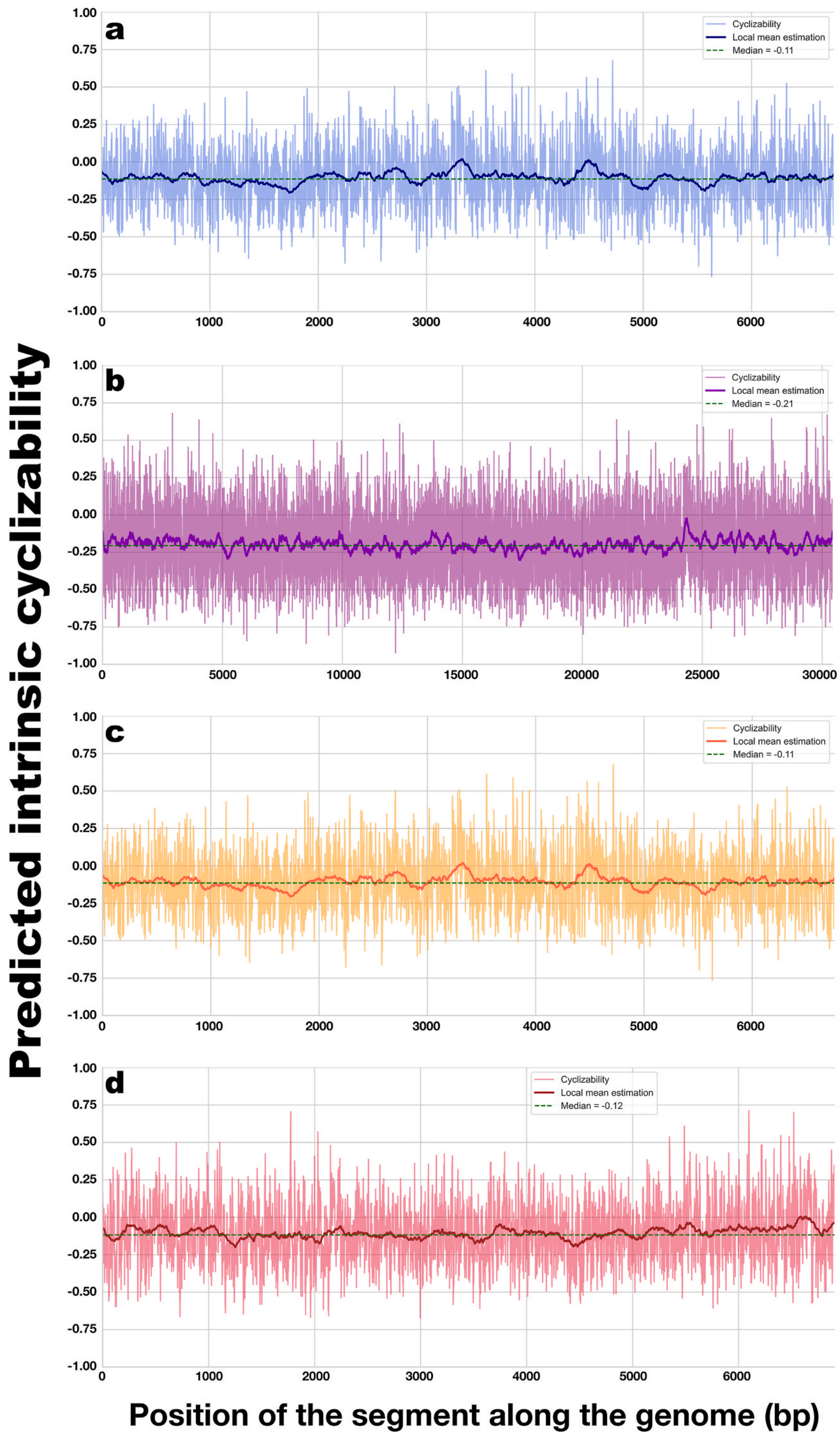

Fig. 11. Predicted intrinsic cyclizabilities along the phage genome at 7 bp resolution, for **(a)** phage PCG-05T, **(b)** phage PCG-06T, **(c)** phage PCG-07T, and **(d)** phage PCG-09T.

sequence identity (Table 2) to published reference genomes genomes (such as Bacillus phages vB_BthS_BMBphi, BC-7, and BC-6) currently precludes their classification into any established viral families or genera. Their low similarity to reference genomes means they currently cannot be classified into existing viral families or genera. They occupy isolated branches on the viral proteomic tree, firmly establishing them as phylogenetically distinct from mainstream phage isolates. Furthermore, comparative network analyses reveal a fascinating ecological dynamic: despite being isolated against the same *Pseudomonas coronafaciens* pv. *garcae* host, these phages diverge into two completely distinct evolutionary lineages (Figs. 2 and 3). Phages PCG-05T, PCG-07T, and PCG-09T form a unique cluster and show distant relatedness to phages targeting Listeria, Carnobacterium, and Enterococcus, whereas phage PCG-06T occupies a solitary phylogenetic branch, sharing structural colinearity only with Bacillus phage 0305phi8-36 (Fig. 2). Collectively, these genomic signatures confirm the isolation of highly unique, previously uncharacterized lytic viral entities that significantly expand our current understanding of bacteriophage diversity. Hence, the lack of recognizable sequence homology with established viral families, the divergence into two unrelated lineages despite targeting the same host

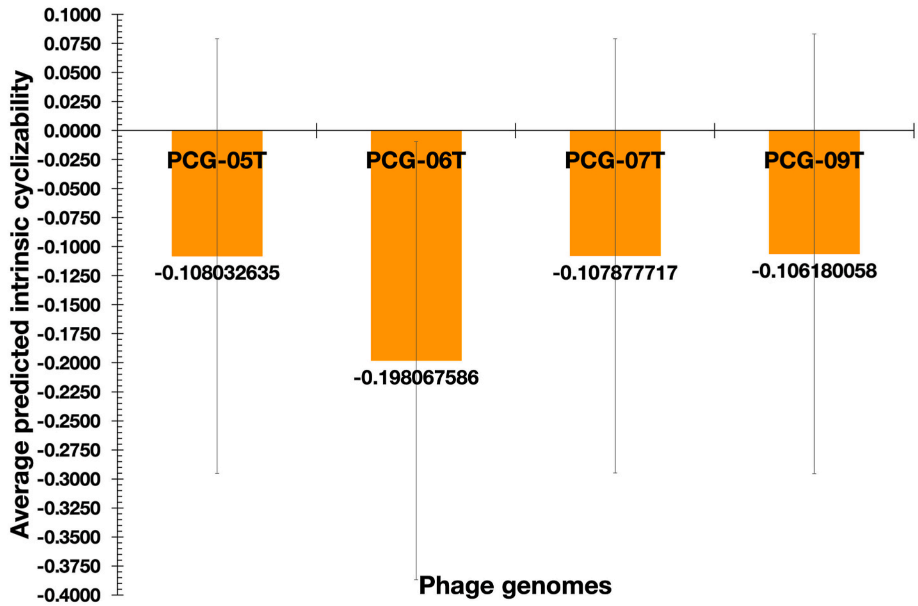

Fig. 12. Average predicted intrinsic cyclizabilities of phages PCG-05T, PCG-06T, PCG-07T and PCG-09T, and associated standard deviations.

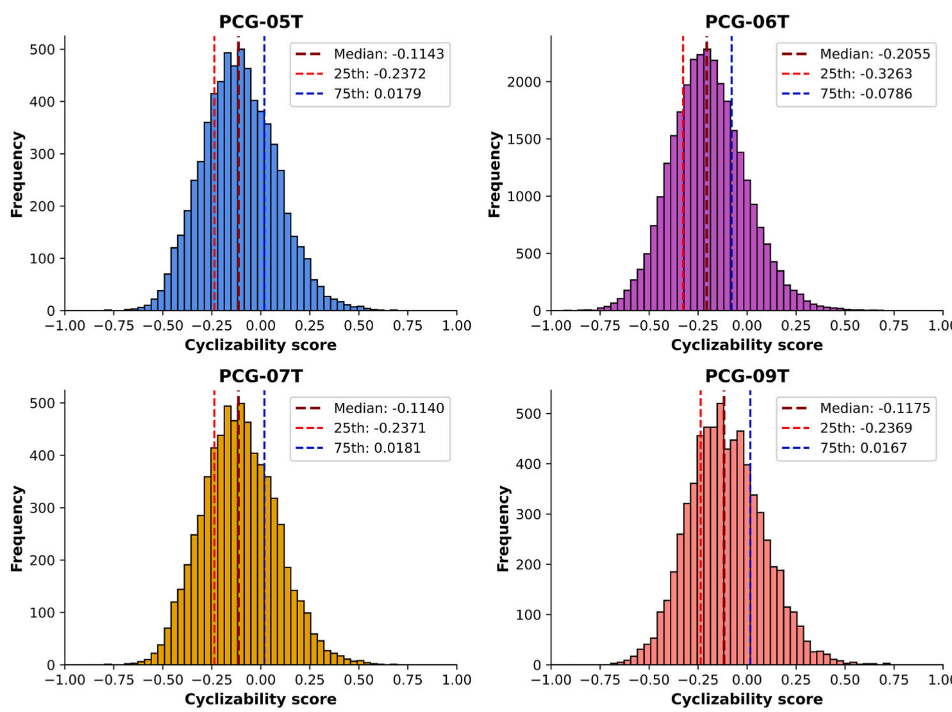

Fig. 13. Distribution of cyclizability values and frequency of apparition in the PCG-05T, PCG-06T, PCG-07T and PCG-09T phage genomes.

Table 4
Statistics of the predicted intrinsic cyclizability data of the genomes of phages PCG-05T, PCG-06T, PCG-07T and PCG-09T.

Groups	Cyclizability value count	Sum	Average	Standard deviation	Variance	Median cyclizability
PCG-05T	6758	−730.0845487	−0.108032635	0.187089699	0.035002555	−0.11429109
PCG-06T	30438	−6028.781185	−0.198067586	0.188662218	0.035593433	−0.20553909
PCG-07T	6758	−729.0376107	−0.107877717	0.187078928	0.034998525	−0.11397403
PCG-09T	6905	−733.1733009	−0.106180058	0.189447914	0.035890512	−0.11751416

species, and the heavy burden of hypothetical "dark matter" proteins collectively cement the uniqueness of these four bacteriophages. In addition, an indepth analysis of the phages' lifestyle using the computational tool BACPHLIP (Hockenberry and Wilke, 2021) provided a more robust phage lifestyle prediction. Hence, there is strong evidence for all four phages following a virulent lifestyle, with stronger evidence for phages PCG-06T and PCG-09T.

In phage ecology, it is a well-documented phenomenon to isolate

Table 5
Results from the one-way ANOVA statistical analysis performed to the whole set of predicted intrinsic cyclizability data of phages PCG-05T, PCG-06T, PCG-07T and PCG-09T genomes.

Source of Variation	SS	df	MS	*F*-ratio	*p*-value	*F*-critical
Between Groups	100.583	3.000	33.528	945.072	0.000000	2.605
Within Groups	1804.143	50855	0.035			
Total	1904.726	50858				

Notes: *SS*: sum of squares; *df*: degrees of freedom; *MS*: mean square (variance estimate; MS=SS/df); *F-ratio*: $MS_{between}/MS_{within}$; *p*-value: probability that the mean will be ≥ (or ≤) than observed results, given that the null hypothesis is true; *F-critical*: statistical *F*-value (1; 45197; 0.05).

identical or near-identical viral genotypes from spatially or temporally separate environments due to stable environmental persistence or clonal expansion. The matrix in Table 2 reveals that phage PCG-09T is a closely related sister species sharing 90.73% ANI and 86.18% AAI with the PCG-05T/PCG-07T lineage, while phage PCG-06T shares 0.00% identity with the rest, highlighting its status as an entirely unique and unrelated phage group. Regarding phages PCG-05T and PCG-07T, we categorize them as distinct isolates (or distinct biological strains) rather than duplicate entries because they were isolated from completely independent sampling events or environments and using different bacterial host strains. The clonal identity of phages PCG-05T and PCG-07T thus frames them as independent spatial/temporal environmental isolates of the same viral species.

DNA shape provides structural insights into the genome that extend beyond the linear nucleotide sequence (Parker and Tullius, 2011). Since DNA function is governed not only by its sequence but also by its three-dimensional structure, variations such as bends, grooves, and twists can influence the way proteins and other molecules interact with the genome (Parker and Tullius, 2011; Yesudhas et al., 2017). These structural features play a critical role in numerous regulatory processes (Rohs et al., 2009). Some of the DNA shape parameters are the minor groove width (MGW), helical twist, propeller twist (ProT), and roll angle (Roll), which collectively capture the geometric characteristics of the DNA helix.

Helical twist (HelT) is the angular rotation between adjacent base pairs, describing how much one base pair is twisted relative to the next. HelT is associated with DNA flexibility and curvature (Dickerson and Klug, 1983; Kriegel et al., 2018). The minor groove width (MGW) is the distance between the sugar-phosphate backbones on the minor groove side of the DNA helix and plays a crucial role in protein-DNA recognition (Laughton and Luisi, 1999; Luscombe et al., 2000; Stella et al., 2010). Propeller twist (ProT) is the rotation about the long axis of the DNA helix, producing non-coplanar bases within a pair, which enhances base stacking and contributes to base pair stability (El Hassan and Calladine, 1996). Roll is the angular bending between base pairs into the grooves of DNA and is an important local parameter for assessing DNA curvature (Tolstorukov et al., 2007).

In this study, using *DNAshape,* a high-throughput computational method that predicts DNA structural features directly from nucleotide sequences (Zhou et al., 2015), we computed the DNA shape parameters for the PCG-05T, PCG-06T, PCG-07T, and PCG-09T phage genomes individually.

Our analysis reveals that phage PCG-05T exhibits a mean HelT value of 34.420, a mean Roll of −0.899, a mean MGW of 5.010, and a mean ProT of −7.421 (Fig. 4). PCG-06T shows higher values, with a mean HelT of 34.456, Roll of −0.693, MGW of 5.047, and ProT of −7.579 (Fig. 4). PCG-07T shares nearly identical DNA shape characteristics with PCG-05T, displaying the same HelT, Roll, MGW, and ProT values (Fig. 4), indicating a highly similar structural profile. PCG-09T presents a mean HelT of 34.411, Roll of −0.893, MGW of 5.014, and ProT of −7.404 (Fig. 4). On average, all genomes demonstrate a similar global helical conformation. The negative Roll values suggest bending toward the minor groove, while the shared negative ProT values across the genomes imply enhanced base stacking (Fig. 5, ProT).

Smaller ProT (Propeller Twist) values facilitate the sliding of base pairs over one another within the constraints of the DNA backbone, thereby indicating increased structural flexibility (El Hassan and Calladine, 1996). Among the genomes analyzed, PCG-06T exhibits the highest (in module) mean ProT value and a broader distribution of deviations from the mean (Fig. 5, ProT), suggesting a greater degree of structural rigidity relative to the other genomes. In contrast, PCG-09T displays the lowest (in module) mean ProT value, indicating enhanced flexibility in specific regions. PCG-05T and PCG-07T show consistent mean ProT values and distribution profiles, suggesting similar structural characteristics (Fig. 5, ProT). A Mann–Whitney *U* test comparing ProT values across samples reveals that PCG-06T is the only phage genome with statistically significant differences relative to the other phage genomes, underscoring its distinct structural profile.

When comparing Roll angle values, we observed that PCG-05T and PCG-07T exhibit the most negative Roll values, followed by PCG-09T, and lastly PCG-06T (Fig. 4). More negative Roll values are typically associated with greater structural deformation, indicating increased local flexibility. The pronounced negative Roll angles in PCG-05T and PCG-07T may reflect enhanced bending toward the minor groove in specific genomic regions, as illustrated in Fig. 5, Roll. This structural tendency aligns with a more flexible DNA conformation. In contrast, PCG-06T displays the least negative Roll angle (Fig. 4), suggesting reduced minor groove bending and, consequently, decreased local flexibility. These findings are consistent with the corresponding ProT values and were further supported by a Mann–Whitney *U* test, which confirmed a statistically significant difference in Roll angles between PCG-06T and the other phage genomes.

Minor groove width (MGW) is a key structural feature that can indicate local DNA flexibility and plays a critical role in DNA shape recognition by proteins (Wing et al., 1980). A narrower minor groove is generally associated with greater accessibility of base edges to DNA-binding proteins, such as transcription factors (Zhou et al., 2015). This increased accessibility may result from compression of the groove, often caused by the close spacing of recognition helices in helix-turn-helix motifs (Kostrewa et al., 1991). Oguey and colleagues (Oguey et al., 2010) further demonstrated that the relationship between MGW and flexibility is likely an intrinsic property of DNA. In our study, phage PCG-06T genome displayed the widest mean MGW value among the genomes, suggesting slightly reduced flexibility in local regions of the minor groove, as visualized in Fig. 5, MGW. In contrast, phages PCG-05T and PCG-07T genomes exhibited the narrowest MGW values, followed closely by phage PCG-09T genome, implying potentially greater accessibility and local flexibility in these genomes (Fig. 5, MGW). A Mann-Whitney test shows a statistical difference between phage PCG-06T genome and the other phage genomes, suggesting a distinct distribution of MGW values when compared to the other genomes.

A Spearman correlation analysis was conducted to examine the relationships among DNA shape features within each phage genome independently. The results obtained are visualized in a series of phage genome-specific plots, displaying both correlation coefficients and significance levels (Fig. 6).

A strong positive correlation was consistently observed between Roll angle and Minor Groove Width (MGW) across all genomes, with a mean Spearman coefficient of 0.53. The highest correlation was found in PCG-

09T (0.55), and the lowest in PCG-06T (0.51) (Fig. 6). This finding aligns with previous work by Fratini and colleagues (Fratini et al., 1982), which demonstrated that Roll reflects the degree to which successive base pairs open toward the minor groove (thus, larger (more positive) Roll angles are often associated with wider MGWs). Conversely, a negative correlation between Helical Twist (HelT) and MGW was observed across all genomes, with a mean coefficient of −0.23. This supports earlier findings by Liebl and colleagues (Liebl et al., 2015), who showed that DNA overwinding, which increases Helical Twist, compresses the minor groove, leading to a narrower MGW. Therefore, this inverse relationship likely reflects a structural trade-off between twisting and groove accessibility.

Additionally, we observed a consistent negative correlation between Propeller Twist (ProT) and Helical Twist in all phage genomes (Fig. 6). Propeller twisting optimizes base stacking, and values approaching 36° tend toward zero, indicating optimal stacking interactions. Deviations from this state can weaken stacking. El Hassan and Calladine (El Hassan and Calladine, 1996) noted that more negative ProT values in double-stranded DNA are associated with increased flexibility, whereas ProT values near −1 to −3° correspond to more rigid DNA structures. This is further supported by Basu and colleagues (Basu et al., 2022), who showed that higher ProT values are associated with lower cyclizability, indicating reduced DNA bendability.

Our results suggest that all phage genomes exhibit correlations among DNA structural features, which may influence their mechanical properties. These DNA shape characteristics (such as narrow minor groove widths, distinct helical twist conformations, variations in Roll, and fluctuations in Propeller Twist) could collectively impact the phage virions biological properties. Specifically, such structural features may facilitate more efficient DNA–protein interactions, enhance accessibility, and contribute to the structural stability of DNA during biologically critical processes like translocation.

While our findings suggest a potentially higher degree of rigidity in specific regions of PCG-06T compared to PCG-05T, PCG-07T, and PCG-09T, it is important to emphasize that these analyses primarily assess DNA structure at the local level. As such, the broader implications of these localized patterns on genome-wide architecture and function remain uncertain. Further studies are warranted to validate these observations and to explore their significance within a global structural and functional context.

Cyclizability, pairwise distant distribution function, and dinucleotide frequency were also evaluated. Cyclizability refers to the propensity of a DNA molecule to undergo cyclization, serving as a proxy for its bendability. The concept was first explored by Shore and Baldwin (1983), who demonstrated the helical nature of DNA based on the periodicity observed in cyclization efficiency (Podtelezhnikov et al., 2000). Since then, numerous experimental and low-throughput approaches have been developed to study DNA cyclization, given its relevance to chromatin organization and transcriptional regulation. This intrinsic property of DNA influences how genomic material is packed and accessed within the nucleus, playing a crucial role in gene expression and genome function.

Basu and colleagues (Basu et al., 2021b) introduced a high-throughput method called *loop-seq* to quantify the intrinsic cyclizability of DNA. This technique estimates cyclizability by computing the logarithm of the ratio between the probability of sequences found in the looped group and those in a control group. In addition, the intrinsic cyclizability was defined as the average value obtained from multiple independent *loop-seq* measurements, providing a robust metric of DNA bendability.

In our study, we independently assessed the intrinsic cyclizability of phages PCG-05T, PCG-06T, PCG-07T, and PCG-09T genomes with a resolution of 7 base pairs, following the methodology described by Basu and colleagues (Basu et al., 2022). Their approach consisted of evaluating cyclizability across a comprehensive library of approximately 90, 000 unique 50-base-pair DNA sequences, enabling high-resolution mapping of DNA bendability.

Our analysis revealed both similarities and differences in the intrinsic cyclizability profiles of the four phage genomes. To visualize these patterns, we have generated heatmaps (Fig. 10), histograms (Fig. 13), and plots highlighting the local mean estimation of the intrinsic cyclizability (Fig. 11), which collectively provided a detailed view of cyclizability across genomic positions (Figs. 10 and 11).

Fig. 11 shows that the median intrinsic cyclizability values differ across the phage genomes, with values of −0.1143 for PCG-05T, −0.2055 for PCG-06T, −0.1140 for PCG-07T, and −0.1175 for PCG-09T. Although the differences between median values appear small, a Mann–Whitney U test revealed a statistically significant difference between PCG-05T and PCG-06T, but not between PCG-05T and either PCG-07T or PCG-09T. Additionally, PCG-06T was significantly different from both PCG-07T and PCG-09T, while no statistical difference was observed between PCG-07T and PCG-09T. These results suggest that the intrinsic cyclizability of phage PCG-06T genome is distinct, with a more negative median intrinsic cyclizability value of −0.2055 compared to the other genomes, probably due to the presence of both low and high cyclizability peaks along the genome. This distinction is further supported by the distribution and frequency of cyclizability values (Fig. 13), which highlight variability of cyclizability peaks across the genome.

In addition, the distribution of intrinsic cyclizability values for each genome is presented in Fig. 13, with the 25th and 75th percentiles indicated. We observed that, for three of the genomes, the 75th percentile of intrinsic cyclizability values falls within the positive range, indicating the presence of regions with higher cyclizability. In contrast, the 75th percentile for the PCG-06T phage genome remains negative (Fig. 13), suggesting a reduced number of regions with high positive cyclizability. This pattern points to a more rigid nature in certain regions of the PCG-06T phage genome compared to the other phage genomes.

While intrinsic cyclizability reflects the looping potential of DNA molecules, it is important to note that DNA looping predominantly occurs in short base pair sequences due to their inherent flexibility (Basu et al., 2021b). In our analysis, we identified local intrinsic cyclizability values within each genome, revealing specific regions (peaks) characterized by either high or low cyclizability. Although this approach provides valuable insights into local DNA bendability, it does not represent a global mechanical property of the entire genome.

Several studies have demonstrated a link between DNA flexibility/cyclizability and specific dinucleotide steps at the local sequence level. A dinucleotide consists of two adjacent nucleotides, and the presence of particular dinucleotide combinations can influence the geometric and structural properties of DNA (Wang and Yan, 2011). The frequency of dinucleotides is often used as a genomic signature across various organisms (Gu et al., 2019). The expected frequency of a dinucleotide is calculated as the product of the individual nucleotide frequencies; in a random sequence, the observed-to-expected (O/E) ratio is 1 (Karlin and Mrázek, 1997). Ratios greater than 1 indicate over-representation, while those below 1 suggest under-representation (Di Giallonardo et al., 2017). Moreover, when a dinucleotide recurs at regular intervals within a sequence, it indicates periodicity. Wu and colleagues (Wu et al., 2011) reported that certain dinucleotides exhibit ~10 bp periodicity, aligning with the helical pitch of DNA, which is associated with increased DNA flexibility. However, periodicity may also be detected at other repetitive intervals depending on sequence context.

Basu and colleagues (Basu et al., 2022) investigated how the spatial organization of specific dinucleotides influences intrinsic cyclizability at the local level. Using high-throughput data, they computed the pairwise distance distribution function to assess the frequency with which specific dinucleotide steps occur at varying distances. This metric quantifies the normalized occurrence of a given dinucleotide step at a specific distance, relative to its expected frequency in a random one. By analyzing these patterns over 100 base pair steps, the study revealed associations between dinucleotide spatial arrangements and DNA mechanical properties such as cyclizability. In their study, they showed how dinucleotide pairs with all As or Ts make a positive contribution to intrinsic cyclizability. Moreover, they showed a positive contribution of Gs or Cs to cyclizability.

In the study reported herein, we computed the pairwise dinucleotide distance correlation for all four phage genomes individually. Although the analysis encompassed all 256 possible dinucleotide pairs, we focused specifically on those previously identified by Basu and colleagues (Basu et al., 2022) as positively contributing to cyclizability: AA-AA, AA-AT, AA-TA, AA-TT, TA-TA, TT-AT, TT-TA, TT-TT, GC-GC, GC-CG, GG-CG, and GG-GC.

We observed that the AA-AA dinucleotide pair exhibited the highest distance correlation in the PCG-09T phage genome, with a correlation frequency value of approximately 1.5, indicating over-representation relative to a random sequence (Fig. 8, AA-AA). In contrast, the same dinucleotide pair in the other genomes showed a distance correlation frequency of 1 or below, suggesting under-representation or random occurrence (Fig. 8, AA-AA).

The AA-AT dinucleotide, which has been associated with a narrow minor groove width in both free and protein-bound DNA (Oguey et al., 2010), showed a correlation frequency of 1 for the PCG-09T and PCG-06T phage genomes and under-representation for the PCG-05T and PCG-07T phage genomes with a correlation frequency of approximately 0.5 (Fig. 8, AA-AT). The AA-TT dinucleotide exhibited a distance correlation frequency of 1 in all phage genomes, suggesting neither over- nor under-representation (Fig. 8, AA-TT). In contrast, the AA-TA dinucleotide was overrepresented in the PCG-09T phage genome, with a distance correlation frequency of approximately 1.3, whereas the remaining phage genomes showed frequencies below 1, indicating under-representation (Fig. 8, AA-TA). The TA-TA dinucleotide shows to be under-represented in all four phage genomes (Fig. 8, TA-TA). The TT-AT dinucleotide was underrepresented in the PCG-09T phage genome and showed no over-representation in the other phage genomes, maintaining a distance correlation frequency near 1 (Fig. 8, TT-AT).

The TT-TA dinucleotide was underrepresented in both PCG-09T and PCG-06T phage genomes but overrepresented in PCG-05T and PCG-07T phage genomes with a similar correlation frequency of approximately 1.3 (Fig. 8, TT-TA). Similarly, the TT-TT dinucleotide pair was overrepresented in the PCG-05T and PCG-07T phage genomes, approximately equal to the random expectation in the PCG-06T phage genome and underrepresented in the PCG-09T phage genome (Fig. 8, TT-TT).

The GC-GC dinucleotide pair exhibited a more irregular distance correlation frequency pattern, with multiple peaks above 1 observed in the PCG-06T and PCG-07T phage genomes, suggesting localized over-representation but no consistency across the genome (Fig. 8, GC-GC). The GG-GC dinucleotide pair showed a distance correlation frequency of approximately 1.3 in the PCG-09T phage genome, indicating over-representation, while all other phage genomes displayed under-representation for this dinucleotide pair (Fig. 8, GG-GC). The GG-CG dinucleotide was not overrepresented in any phage genome and was underrepresented specifically in the PCG-05T and PCG-07T phage genomes (Fig. 8, GG-CG). Lastly, the GC-CG dinucleotide pair was underrepresented in three of the phage genomes and aligned with the expected random frequency only in the PCG-06T phage genome (Fig. 8, GC-CG).

As observed, the pairwise distance correlation frequencies highlight the over- and under-representation of dinucleotides previously identified by Basu and colleagues (Basu et al., 2022) as positively associated with cyclizability. These patterns reveal a notable variability across phage genomes. For instance, AA's dinucleotides such as AA-AA and AA-TA are overrepresented in the PCG-09T phage genome, while AA-AT remains randomly represented. In contrast, several TT's dinucleotides (such as TT-AT, TT-TA, and TT-TT) are underrepresented in the PCG-09T phage genome but show over-representation in the PCG-05T and PCG-07T phage genomes (Fig. 8).

Additionally, the PCG-06T phage genome demonstrates over-representation in GG-GC but under-representation in GC-CG (Fig. 8). These findings suggest that the presence and spatial arrangement of specific dinucleotides contribute differently to intrinsic cyclizability across phage genomes. However, further experimental validation is necessary to substantiate these associations and fully understand their functional implications. Moreover, no special distribution of peaks was observed in either of the dinucleotides and phage genomes.

In addition, we analyzed the distribution of all 16 possible dinucleotides (AA, AC, AG, AT, CA, CC, CG, CT, GA, GC, GG, GT, TA, TC, TG, and TT) across all phage genomes by calculating their frequency of occurrence. The complete distribution of dinucleotide frequencies for each genome is shown in Fig. 7. The most frequent dinucleotide in the PCG-05T phage genome was CT, accounting for 8.069% of all dinucleotides. In PCG-06T, AT was the most abundant at 8.615%, while CT was again most frequent in PCG-07T at 8.069%. For PCG-09T, TA showed the highest occurrence at 8.166%.

Basu and colleagues (Basu et al., 2022) reported that the TA dinucleotide had the strongest positive influence on intrinsic cyclizability, whereas CG was identified as the least flexible. Earlier studies (Packer et al., 2000) demonstrated that TA dinucleotides impart unique mechanical properties to DNA, including enhanced flexibility. These findings were further supported by a more recent work (Back and Walther, 2023), who confirmed a strong correlation between TA dinucleotide presence and DNA cyclizability, reinforcing its role in promoting DNA flexibility.

Conversely, CG dinucleotides are associated with reduced flexibility, particularly due to their susceptibility to cytosine methylation. Rao and colleagues (Rao et al., 2018) found that CG sites are the most frequent targets of methylation, a modification that alters local DNA structure and shape (especially Roll and ProT). Their study showed that methylated CpG steps exhibited increased Roll angles and decreased ProT values, with effects accounting for 50–100% of the range observed in unmethylated sequences. Basu and colleagues (Basu et al., 2022) further confirmed that CpG methylation leads to increased DNA rigidity within individual sequences.

While the pairwise dinucleotide distance correlation analysis did not reveal over- or clear under-representation of the TA dinucleotide in any of the phage genomes, the frequency of occurrence analysis showed that TA is the most abundant dinucleotide in the PCG-09T phage genome. This elevated occurrence suggests a higher potential for DNA flexibility in this phage genome compared to the other phage genomes. Nonetheless, the TA dinucleotide also occurs at relatively high frequencies in the other phage genomes—7.986% in PCG-05T, 7.305% in PCG-06T, and 7.989% in PCG-07T—suggesting a degree of flexibility for these genomes in local areas along the genomes (Fig. 7).

The CG dinucleotide exhibits one of the lowest frequencies across the phage genomes, with values of 2.872% in PCG-05T, 4.830% in PCG-06T, 2.872% in PCG-07T, and 2.844% in PCG-09T (Fig. 7). Notably, the PCG-09T phage genome has the lowest CG content, further supporting its relatively high(er) intrinsic cyclizability compared to the other phage genomes (Fig. 12). Analysis of the pairwise distance distribution function shows that the GC-CG dinucleotide is underrepresented in all genomes except for PCG-06T (Fig. 8, GC-CG), where it exceeds the random expectation with a distance correlation frequency of approximately 1.3. This elevated correlation may suggest increased structural rigidity in some areas of the PCG-06T genome, correlating with the greatest negative intrinsic cyclizability found in this genome (Fig. 12); however, further investigation is needed to confirm this hypothesis.

To better have a picture of the effect and difference of the frequency of the dinucleotides in each phage genome, we calculated the differential dinucleotide frequency. The results obtained show 256 frequency values that were depicted in a Figure heatmap (Fig. 9). In this heatmap, red signifies a high difference, indicating that certain dinucleotides are more represented compared to other dinucleotides from the other genome. Conversely, the blue color indicates a low difference, suggesting that these specific dinucleotides are less represented than other dinucleotides from the other genome (Fig. 9).

Our analysis revealed a higher differential dinucleotide frequency (particularly for CC, CG, GC, and GG) across the genomes, probably due to the low frequency of occurrence and likely reflecting differences in the underlying nucleotide composition. Notably, we observed strong similarity between PCG-05T and PCG-07T phage genomes, reinforcing their molecular compositional resemblance. In addition, the PCG-06T phage genome displayed a distinct dinucleotide profile, particularly with the CT dinucleotide, which was significantly lower in frequency compared to the other phage genomes where it remained relatively consistent. This reduction may suggest a selective usage or avoidance of this dinucleotide in the PCG-06T phage genome. Additionally, we showed a higher differential frequency of the CG dinucleotide in PCG-09T, further supporting its unique genomic features. The elevated differential frequencies of these dinucleotides across the phage genomes suggest a potential role in modulating DNA structure and influencing local flexibility, either by enhancing or restricting DNA bendability.

According to a mechanistic rationale deployed by Balcão and colleagues (Balcão et al., 2022), dsDNA sequences rich in GC dinucleotides are less flexible (i.e., more rigid) than sequences rich in AT dinucleotides. Despite having a higher net 41.8% GC (and 58.2% AT) content in its genome (Table 3), phage PCG-06T dsDNA sequence is in fact rich(er) in GC dinucleotides (Fig. 7) with 4.191% (thus imparting a more rigid genome than the genome of phage PCG-09T with 3.472% (Fig. 7), which is in clear agreement with the (much lower) value obtained for the virion morphogenesis yield of phage PCG-06T. Hence, the genome of phage PCG-09T is apparently more flexible than the genome of phage PCG-06T, which is in clear agreement with the values obtained for the virion morphogenesis yield of both phages. These observations are consistent with the rationale that upon translocation of a more bendable dsDNA genome into the bacterial host cytoplasm, the bacterial host RNA polymerase interacts and negotiates with it in a much more efficient way, leading to higher transcription rates and production of phage proteins in larger numbers with concomitant assembly of a larger number of mature virions (Balcão et al., 2022).

Finally, it is important to emphasize that the properties analyzed in this study (DNA shape, pairwise distance correlations, and intrinsic cyclizability) primarily reflect local or nucleotide-level features of the genome. While these characteristics provide valuable insights into local DNA flexibility and structural dynamics, they are not solely determinative of genome-wide behavior, as broader structural and epigenetic factors also play significant roles at the global level. These additional factors were beyond the scope of the current analysis. Nonetheless, our study offers a detailed nucleotide-level characterization of PCG-05T, PCG-06T, PCG-07T and PCG-09T phage genomes and sheds light on their intrinsic flexibility and cyclizability. We acknowledge that a comprehensive global analysis would offer a more complete understanding of these phage genomes and plan to address this in future work.

Biological implications of genomic and biophysical features. The biophysical and genomic features reported here are central to the biological performance of these phages as biocontrol agents. The comprehensive suite of DNA replication and repair machinery identified in our annotation (e.g., DNA polymerases, helicases, and resolvases) is not merely descriptive; it signifies the phage's capacity for rapid, efficient hijacking of the *Pseudomonas coronafaciens* pv. *garcae* host machinery. This metabolic autonomy is directly linked to the short latent periods and high burst sizes required to suppress bacterial populations in agricultural settings. Furthermore, the stability of the phage particles, supported by the presence of a robust structural gene module (head and tail proteins), provides the physical resilience necessary for survival in fluctuating environmental conditions (e.g., UV exposure or temperature variations on leaf surfaces, unpublished data). By establishing that these phages possess the necessary genetic toolkit for rapid replication without carrying deleterious virulence or antibiotic resistance factors, we demonstrate that their physical structure is optimized for both safety and biological efficacy. This ensures that the phages can successfully complete their infection cycle (from initial attachment to host lysis) without negatively impacting the crop or the broader environment.

5. Conclusions

The findings reported in the present work not only expand the catalog of phages available for biocontrol of coffee plant-associated phytopathogens but also highlight the relevance of integrating genomic mechanics into our understanding of phage biology.

The results presented herein clearly suggest that different phage DNA shapes influence their mechanical (flexibility) properties which, in turn, might play an effective role in the virion morphogenesis yield upon translocation of phage DNA into the bacterial cytoplasm following contraction of the virion tail sheath. Hence, a more flexible phage genome (such as that of phage PCG-09T) apparently leads to a higher virion morphogenesis yield upon successful infection of a susceptible bacterial host cell, and in turn, a less flexible phage genome (such as that of phage PCG-06T) apparently leads to a lower virion morphogenesis yield.

These conclusions are fully backed up by findings from Basu and colleagues (Basu et al., 2021a, 2021b, 2022) and later confirmed by Balcão and colleagues (Balcão et al., 2022), that dsDNA sequences rich in the dinucleotide TA display much higher intrinsic cyclizability values and are therefore endowed with a higher bendability which, in turn, is consistent with the mechanistic rationale that the bacterial host cell RNA-polymerase might interact and negotiate better with such dsDNA leading to higher transcription rates with concomitant higher levels of protein synthesis and virion morphogenesis yields.

While our findings suggest a potentially higher degree of rigidity in specific regions of PCG-06T compared to PCG-05T, PCG-07T, and PCG-09T, it is important to emphasize that these analyses primarily assess DNA structure at the local level. As such, the broader implications of these localized patterns on genome-wide architecture and function remain uncertain. Further studies are warranted to validate these observations and to explore their significance within a global structural and functional context.

CRediT authorship contribution statement

Fernanda C. Moreli: Visualization, Methodology, Investigation. **Carlos A. Quinde:** Validation, Software, Methodology, Formal analysis, Data curation. **Basilio Cieza:** Visualization, Validation, Supervision, Software, Methodology, Formal analysis, Data curation. **Aakash Basu:** Software, Methodology, Formal analysis. **Marta M.D.C. Vila:** Writing – review & editing, Visualization, Validation, Resources, Formal analysis. **Carla Pereira:** Writing – review & editing, Visualization, Validation, Formal analysis. **Paul Hyman:** Writing – review & editing, Writing – original draft, Visualization, Validation, Supervision, Formal analysis. **Victor M. Balcão:** Writing – review & editing, Writing – original draft, Visualization, Validation, Supervision, Resources, Project administration, Methodology, Funding acquisition, Formal analysis, Data curation, Conceptualization.

Ethics committee authorization

Not applicable.

Funding

This research was funded by São Paulo Research Foundation (FAPESP), grant 2022/10775-9 (Project PsgPhageKill). V.M.B. received a research fellowship award from the National Council for Scientific and Technological Development (CNPq) (grant 301978/2022-0). Fernanda Moreli was supported by a Ph.D. fellowship from the Coordination for the Improvement of Higher Education Personnel (CAPES).

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgements

The authors are grateful to Prof. Paul Hyman (Department of Biology/Toxicology, Ashland University, Ohio OH 44805, U.S.A.) for his guidance regarding the discussion of the polyvalency of the four Pcg phages that entered the study. The authors are also grateful to Maria Isabel Scarpa de Arruda (“Bebel”), for granting unlimited access to the 50-year-old coffee plants at her coffee plantation (Fazenda Santo Antônio da Bela Vista (Itu/SP, Brazil)). Thanks are also due to Dr. Horacio Montenegro, at NGS, for his help with the bioinformatic analyses.

Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.phage.2026.100010.

Data availability

Data will be made available on request.

References

Abe, N., Dror, I., Yang, L., Slattery, M., Zhou, T., Bussemaker, H.J., Rohs, R., Mann, R.S., 2015. Deconvolving the recognition of DNA shape from sequence. Cell 161, 307–318. https://doi.org/10.1016/j.cell.2015.02.008.

Alexandrov, L.B., Bishop, A.R., Rasmussen, K.Ø., Alexandrov, B.S., 2016. The role of structural parameters in DNA cyclization. BMC Bioinf. 17, 68. https://doi.org/10.1186/s12859-016-0897-9.

Back, G., Walther, D., 2023. Predictions of DNA mechanical properties at a genomic scale reveal potentially new functional roles of DNA flexibility. NAR Genom. Bioinf. 5. https://doi.org/10.1093/nargab/lqad097 lqad097.

Balcão, V.M., Basu, A., Cieza, B., Rossi, F.N., Pereira, C., Vila, M.M.D.C., Setubal, J.C., Ha, T., da Silva, A.M., 2022. Pseudomonas-tailed lytic phages: genome mechanical analysis and putative correlation with virion morphogenesis yield. Future Microbiol. 17, 1009–1026. https://doi.org/10.2217/fmb-2021-0293.

Basu, A., Bobrovnikov, D.G., Cieza, B., Arcon, J.P., Qureshi, Z., Orozco, M., Ha, T., 2022. Deciphering the mechanical code of the genome and epigenome. Nat. Struct. Mol. Biol. 29, 1178–1187. https://doi.org/10.1038/s41594-022-00877-6.

Basu, A., Bobrovnikov, D.G., Ha, T., 2021a. DNA mechanics and its biological impact. J. Mol. Biol. 433, 166861. https://doi.org/10.1016/j.jmb.2021.166861.

Basu, A., Bobrovnikov, D.G., Qureshi, Z., Kayikcioglu, T., Ngo, T.T.M., Ranjan, A., Eustermann, S., Cieza, B., Morgan, M.T., Hejna, M., Rube, H.T., Hopfner, K.-P., Wolberger, C., Song, J.S., Ha, T., 2021b. Measuring DNA mechanics on the genome scale. Nature 589, 462–467. https://doi.org/10.1038/s41586-020-03052-3.

Belan, L.L., Pozza, E.A., Freitas, M.L.O., Raimundi, M.K., Souza, R.M., Machado, J.C., 2016. Occurrence of Pseudomonas syringae pv. garcae in coffee seeds. Aust. J. Crop. Sci. 10, 1015–1021. https://doi.org/10.21475/ajcs.2016.10.07.p7693.

Chakraborty, S., Newton, A.C., 2011. Climate change, plant diseases and food security: an overview. Plant Pathol. 60, 2–14. https://doi.org/10.1111/j.1365-3059.2010.02411.x.

Cohen, R.S., Meselson, M., 1988. Periodic interactions of heat shock transcriptional elements. Nature 332, 856–858. https://doi.org/10.1038/332856a0.

Crothers, D.M., Drak, J., Kahn, J.D., Levene, S.D., 1992. DNA bending, flexibility, and helical repeat by cyclization kinetics. Methods Enzymol. 212, 3–29. https://doi.org/10.1016/0076-6879(92)12003-9.

de Souza, J.O.G., Pozza, E.A., Belan, L.L., Freitas, M.L.O., Oliveira, J.M., 2019. Initial stages of infectious process of Pseudomonas syringae pv. garcae in coffee leaves. Coffee Sci. Lavras 14, 338–348. https://doi.org/10.25186/cs.v14i3.1585.

Di Giallonardo, F., Schlub, T.E., Shi, M., Holmes, E.C., 2017. Dinucleotide composition in animal RNA viruses is shaped more by virus family than by host species. J. Virol. 91. https://doi.org/10.1128/JVI.02381-16 e02381-16.

Dickerson, R.E., 1998. DNA bending: the prevalence of kinkiness and the virtues of normality. Nucleic Acids Res. 26, 1906–1926. https://doi.org/10.1093/nar/26.8.1906.

Dickerson, R.E., Klug, A., 1983. Base sequence and helix structure variation in B and A DNA. J. Mol. Biol. 166, 419–441. https://doi.org/10.1016/s0022-2836(83)80093-x.

El Hassan, M.A., Calladine, C.R., 1996. Propeller-twisting of base-pairs and the conformational mobility of dinucleotide steps in DNA. J. Mol. Biol. 259, 95–103. https://doi.org/10.1006/jmbi.1996.0304.

Frampton, R.A., Pitman, A.R., Fineran, P.C., 2012. Advances in bacteriophage-mediated control of plant pathogens. Internet J. Microbiol. 2012, 326452. https://doi.org/10.1155/2012/326452.

Frampton, R.A., Taylor, C., Holguín Moreno, A.V., Visnovsky, S.B., Petty, N.K., Pitman, A.R., Fineran, P.C., 2014. Identification of bacteriophages for biocontrol of the kiwifruit canker phytopathogen Pseudomonas syringae pv. actinidiae. Appl. Environ. Microbiol. 80, 2216–2228. https://doi.org/10.1128/aem.00062-14.

Fratini, A.V., Kopka, M.L., Drew, H.R., Dickerson, R.E., 1982. Reversible bending and helix geometry in a B-DNA dodecamer: CGCGAATTBrCGCG. J. Biol. Chem. 257, 14686–14707.

Gu, H., Fan, R.L.Y., Wang, D., Poon, L.L.M., 2019. Dinucleotide evolutionary dynamics in influenza A virus. Virus Evol. 5. https://doi.org/10.1093/ve/vez038 vez038.

Harada, L.K., Silva, E.C., Campos, W.F., Del Fiol, F.S., Vila, M.M.D.C., Dąbrowska, K., Krylov, V.N., Balcão, V.M., 2018. Biotechnological applications of bacteriophages: state of the art. Microbiol. Res. 212–213, 38–58. https://doi.org/10.1016/j.micres.2018.04.007.

Harada, L.K., Silva, E.C., Rossi, F.P.N., Cieza, B., Oliveira, T.J., Pereira, C., Tomazetto, G., Silva, B.B., Squina, F.M., Vila, M.M.D.C., Setubal, J.C., Ha, T., da Silva, A.M., Balcão, V.M., 2022. Characterization and in vitro testing of newly isolated lytic bacteriophages for biocontrol of Pseudomonas aeruginosa. Future Microbiol. 17, 111–141. https://doi.org/10.2217/fmb-2021-0027.

Hockenberry, A.J., Wilke, C.O., 2021. BACPHLIP: predicting bacteriophage lifestyle from conserved protein domains. PeerJ 9, e11396. https://doi.org/10.7717/peerj.11396.

Hyman, P., 2019. Phages for phage therapy: isolation, characterization, and host range breadth. Pharmaceuticals 12, 35. https://doi.org/10.3390/ph12010035.

Iriarte, F.B., Balogh, B., Momol, M.T., Smith, L.M., Wilson, M., Jones, J.B., 2007. Factors affecting survival of bacteriophage on tomato leaf surfaces. Appl. Environ. Microbiol. 73, 1704–1711. https://doi.org/10.1128/AEM.02118-06.

Jones, J.B., Jackson, L.E., Balogh, B., Obradovic, A., Iriarte, F.B., Momol, M.T., 2007. Bacteriophages for plant disease control. Annu. Rev. Phytopathol. 45, 245–262. https://doi.org/10.1146/annurev.phyto.45.062806.094411.

Kakasis, A., Panitsa, G., 2019. Bacteriophage therapy as an alternative treatment for human infections. A comprehensive review. Int. J. Antimicrob. Agents 53, 16–21. https://doi.org/10.1016/j.ijantimicag.2018.09.004.

Kannan, S., Kohlhoff, K., Zacharias, M., 2006. B-DNA under stress: Over- and untwisting of DNA during molecular dynamics simulations. Biophys. J. 91, 2956–2965. https://doi.org/10.1529/biophysj.106.087163.

Karlin, S., Mrázek, J., 1997. Compositional differences within and between eukaryotic genomes. Proc. Natl. Acad. Sci. USA 94, 10227–10232. https://doi.org/10.1073/pnas.94.19.10227.

Kasurinen, J., Spruit, C.M., Wicklund, A., Pajunen, M.I., Skurnik, M., 2021. Screening of bacteriophage encoded toxic proteins with a next generation sequencing-based assay. Viruses 13, 750. https://doi.org/10.3390/v13050750.

Kharina, A.V., Zaika, S.A., Yumyna, Y.M., Zelena, P.P., Kornienko, N.O., Kosenko, Y.A., Polischuk, V.P., 2015. Detection of Proteus mirabilis and Enterobacter cloacae in tomatto and pepper fruits and isolation of their bacteriophages. Наукові доповіді Національного університету біоресурсів і природокористування України 13, 80–93.

Kim, D., Park, S., Chun, J., 2021. Introducing EzAAI: a pipeline for high throughput calculations of prokaryotic average amino acid identity. J. Microbiol. 59, 476–480. https://doi.org/10.1007/s12275-021-1154-0.

Kostrewa, D., Granzin, J., Koch, C., Choe, H.W., Raghunathan, S., Wolf, W., Labahn, J., Kahmann, R., Saenger, W., 1991. Three-dimensional structure of the E. coli DNA-binding protein FIS. Nature 349, 178–180. https://doi.org/10.1038/349178a0.

Kriegel, F., Matek, C., Dršata, T., Kulenkampff, K., Tschirpke, S., Zacharias, M., Lankaš, F., Lipfert, J., 2018. The temperature dependence of the helical twist of DNA. Nucleic Acids Res. 46, 7998–8009. https://doi.org/10.1093/nar/gky599.

Laughton, C., Luisi, B., 1999. The mechanics of minor groove width variation in DNA, and its implications for the accommodation of ligands. J. Mol. Biol. 288, 953–963. https://doi.org/10.1006/jmbi.1999.2733.

Le, T.T.B., 2019. Bacteriophage: a Potential Treatment for Citrus Canker. Texas A&M University.

Liebl, K., Drsata, T., Lankas, F., Lipfert, J., Zacharias, M., 2015. Explaining the striking difference in twist-stretch coupling between DNA and RNA: a comparative molecular dynamics analysis. Nucleic Acids Res. 43, 10143–10156. https://doi.org/10.1093/nar/gkv1028.

Luscombe, N.M., Austin, S.E., Berman, H.M., Thornton, J.M., 2000. An overview of the structures of protein-DNA complexes. Genome Biol. 1. https://doi.org/10.1186/gb-2000-1-1-reviews001. REVIEWS001.

McManus, P.S., Stockwell, V.O., Sundin, G.W., Jones, A.L., 2002. Antibiotic use in plant agriculture. Annu. Rev. Phytopathol. 40, 443–465. https://doi.org/10.1146/annurev.phyto.40.120301.093927.

Mihara, T., Nishimura, Y., Shimizu, Y., Nishiyama, H., Yoshikawa, G., Uehara, H., Hingamp, P., Goto, S., Ogata, H., 2016. Linking virus genomes with host taxonomy. Viruses 8, 66. https://doi.org/10.3390/v8030066.

Moreli, F.C., Rodrigues, L.M.R., Destefano, S.A.L., Vila, M.M.D.C., Hyman, P., Balcão, V. M., 2026. Enhanced suppression of Pseudomonas coronafaciens pv. garcae in in-vitro and ex-planta studies by a four-phage cocktail: biological, genomic and kinetic profiling. Bacteriophage 1, 100005. https://doi.org/10.1016/j.phage.2026.100005.

Mota, L.C., Silva, E.C., Quinde, C.A., Cieza, B., Basu, A., Rodrigues, L.M.R., Vila, M.M.D. C., Balcão, V.M., 2025. Potential of a newly isolated lytic bacteriophage to control Pseudomonas coronafaciens pv. garcae in coffee plants: molecular characterization with in vitro and ex vivo experiments. Enzym. Microb. Technol. 184, 110573. https://doi.org/10.1016/j.enzmictec.2024.110573.

Nishimura, Y., Yoshida, T., Kuronishi, M., Uehara, H., Ogata, H., Goto, S., 2017. ViPTree: the viral proteomic tree server. Bioinformatics 33, 2379–2380. https://doi.org/10.1093/bioinformatics/btx157.

Oguey, C., Foloppe, N., Hartmann, B., 2010. Understanding the sequence-dependence of DNA groove dimensions: implications for DNA interactions. PLoS One 5, e15931. https://doi.org/10.1371/journal.pone.0015931.
Olson, W.K., Gorin, A.A., Lu, X.J., Hock, L.M., Zhurkin, V.B., 1998. DNA sequence-dependent deformability deduced from protein-DNA crystal complexes. Proc. Natl. Acad. Sci. USA 95, 11163–11168. https://doi.org/10.1073/pnas.95.19.11163.
Packer, M.J., Dauncey, M.P., Hunter, C.A., 2000. Sequence-dependent DNA structure: dinucleotide conformational maps. J. Mol. Biol. 295, 71–83. https://doi.org/10.1006/jmbi.1999.3236.
Parker, S.C., Tullius, T.D., 2011. DNA shape, genetic codes, and evolution. Curr. Opin. Struct. Biol. 21, 342–347. https://doi.org/10.1016/j.sbi.2011.03.002.
Pereira, C., Costa, P., Pinheiro, L., Balcão, V.M., Almeida, A., 2021b. Kiwifruit bacterial canker: an integrative view focused on biocontrol strategies. Planta 253, 49. https://doi.org/10.1007/s00425-020-03549-1.
Pereira, Carla, Costa, P., Pinheiro, L., Balcão, V.M., Almeida, A., 2021a. Kiwifruit bacterial canker: an integrative view focused on biocontrol strategies. Planta 253, 49. https://doi.org/10.1007/s00425-020-03549-1.
Pinheiro, L.A.M., Pereira, C., Barreal, M.E., Gallego, P.P., Balcão, V.M., Almeida, A., 2019a. Use of phage ϕ6 to inactivate Pseudomonas syringae pv. actinidiae in kiwifruit plants: in vitro and ex vivo experiments. Appl. Microbiol. Biotechnol. 104, 1319–1330. https://doi.org/10.1007/s00253-019-10301-7.
Pinheiro, L.A.M., Pereira, C., Frazão, C., Balcão, V.M., Almeida, A., 2019b. Efficiency of phage φ6 for biocontrol of Pseudomonas syringae pv. syringae: an in vitro preliminary study. Microorganisms 7, 286. https://doi.org/10.3390/microorganisms7090286.
Podtelezhnikov, A.A., Mao, C., Seeman, N.C., Vologodskii, A., 2000. Multimerization-cyclization of DNA fragments as a method of conformational analysis. Biophys. J. 79, 2692–2704. https://doi.org/10.1016/S0006-3495(00)76507-6.
Rao, S., Chiu, T.P., Kribelbauer, J.F., Mann, R.S., Bussemaker, H.J., Rohs, R., 2018. Systematic prediction of DNA shape changes due to CpG methylation explains epigenetic effects on protein-DNA binding. Epigenetics Chromatin 11, 6. https://doi.org/10.1186/s13072-018-0174-4.
Richmond, T.J., Davey, C.A., 2003. The structure of DNA in the nucleosome core. Nature 423, 145–150. https://doi.org/10.1038/nature01595.
Ristaino, J.B., Anderson, P.K., Bebber, D.P., Brauman, K.A., Cunniffe, N.J., Fedoroff, N. V., Finegold, C., Garrett, K.A., Gilligan, C.A., Jones, C.M., Martin, M.D., MacDonald, G.K., Neenan, P., Records, A., Schmale, D.G., Tateosian, L., Wei, Q., 2021. The persistent threat of emerging plant disease pandemics to global food security. Proc. Natl. Acad. Sci. USA 118, e2022239118. https://doi.org/10.1073/pnas.2022239118.
Rizzo, D.M., Lichtveld, M., Mazet, J.A.K., Togami, E., Miller, S.A., 2021. Plant health and its effects on food safety and security in a one health framework: four case studies. One Health Outlook 3, 6. https://doi.org/10.1186/s42522-021-00038-7.
Rodrigues, L.M.R., Almeida, I.M.G., Patrício, F.R.A., Beriam, L.O.S., Maciel, K.W., Braghini, M.T., Guerreiro Filho, O., 2013. Mancha Aureolada do Cafeeiro Causada Por Pseudomonas Syringae Pv. Boletim técnico IAC, garcae (Instituto Agronômico Campinas).
Rodrigues, L.M.R., Queiroz-Voltan, R.B., Guerreiro-Filho, O., 2015. Anatomical changes on coffee leaves infected by Pseudomonas syringae pv. garcae. Summa Phytopathol. 41, 256–261. https://doi.org/10.1590/0100-5405/2049.
Rodrigues, L.M.R., Sera, G.H., Filho, O.G., Beriam, L.O.S., Almeida, I.M.G., 2017. First report of mixed infection by Pseudomonas syringae pathovars garcae and tabaci on coffee plantations. Plant Prot. 76, 543–549. https://doi.org/10.1590/1678-4499.2016.399.
Rohs, R., West, S.M., Sosinsky, A., Liu, P., Mann, R.S., Honig, B., 2009. The role of DNA shape in protein-DNA recognition. Nature 461, 1248–1253. https://doi.org/10.1038/nature08473.
Savary, S., Willocquet, L., 2020. Modeling the impact of crop diseases on global food security. Annu. Rev. Phytopathol. 58, 313–341. https://doi.org/10.1146/annurev-phyto-010820-012856.
Savary, S., Willocquet, L., Pethybridge, S.J., Esker, P., McRoberts, N., Nelson, A., 2019. The global burden of pathogens and pests on major food crops. Nat. Ecol. Evol. 3, 430–439. https://doi.org/10.1038/s41559-018-0793-y.
Schoch, C.L., Ciufo, S., Domrachev, M., Hotton, C.L., Kannan, S., Khovanskaya, R., Leipe, D., Mcveigh, R., O'Neill, K., Robbertse, B., Sharma, S., Soussov, V., Sullivan, J. P., Sun, L., Turner, S., Karsch-Mizrachi, I., 2020. NCBI taxonomy: a comprehensive update on curation, resources and tools. Database (Oxford) 2020. https://doi.org/10.1093/database/baaa062 baaa062.
Shaw, J., Yu, Y.W., 2023. Fast and robust metagenomic sequence comparison through sparse chaining with skani. Nat. Methods 20, 1661–1665. https://doi.org/10.1038/s41592-023-02018-3.
Shore, D., Baldwin, R.L., 1983. Energetics of DNA twisting. I. Relation between twist and cyclization probability. J. Mol. Biol. 170, 957–981. https://doi.org/10.1016/s0022-2836(83)80198-3.
Shore, D., Langowski, J., Baldwin, R.L., 1981. DNA flexibility studied by covalent closure of short fragments into circles. Proc. Natl. Acad. Sci. USA 78, 4833–4837. https://doi.org/10.1073/pnas.78.8.4833.
Silva, E.C., Quinde, C.A., Cieza, B., Basu, A., Vila, M.M.D.C., Balcão, V.M., 2024. Molecular characterization and genome mechanical features of two newly isolated polyvalent bacteriophages infecting Pseudomonas syringae pv. garcae. Genes 15, 113. https://doi.org/10.3390/genes15010113.
Silva, E.C., Rodrigues, L.M.R., Vila, M.M.D.C., Balcão, V.M., 2023. Newly isolated phages preying on Pseudomonas syringae pv. garcae: in vitro and ex vivo inactivation studies in coffee plant leafs. Enzym. Microb. Technol. 171, 110325. https://doi.org/10.1016/j.enzmictec.2023.110325.
Silva, F.J., Vieira, B.S., Siquieroli, A.C.S., 2019. Biological control of Pseudomonas syringae pv. garcae in coffee crop with Bacillus spp. isolates. Científica, Jaboticabal 47, 364–370. https://doi.org/10.15361/1984-5529.2019v47n4p364-370.
Singh, B.K., Delgado-Baquerizo, M., Egidi, E., Guirado, E., Leach, J.E., Liu, H., Trivedi, P., 2023. Climate change impacts on plant pathogens, food security and paths forward. Nat. Rev. Microbiol. 21, 640–656. https://doi.org/10.1038/s41579-023-00900-7.
Stella, S., Cascio, D., Johnson, R.C., 2010. The shape of the DNA minor groove directs binding by the DNA-bending protein Fis. Genes Dev. 24, 814–826. https://doi.org/10.1101/gad.1900610.
Svircev, A., Roach, D., Castle, A., 2018. Framing the future with bacteriophages in agriculture. Viruses 10, 218. https://doi.org/10.3390/v10050218.
Tang, L., 2021. Sequencing DNA bendability. Nat. Methods 18, 121. https://doi.org/10.1038/s41592-021-01070-1.
Tolstorukov, M.Y., Colasanti, A.V., McCandlish, D.M., Olson, W.K., Zhurkin, V.B., 2007. A novel roll-and-slide mechanism of DNA folding in chromatin: implications for nucleosome positioning. J. Mol. Biol. 371, 725–738. https://doi.org/10.1016/j.jmb.2007.05.048.
Turner, D., Shkoporov, A.N., Lood, C., Millard, A.D., Dutilh, B.E., Alfenas-Zerbini, P., van Zyl, L.J., Aziz, R.K., Oksanen, H.M., Poranen, M.M., Kropinski, A.M., Barylski, J., Brister, J.R., Chanisvili, N., Edwards, R.A., Enault, F., Gillis, A., Knezevic, P., Krupovic, M., Kurtböke, I., Kushkina, A., Lavigne, R., Lehman, S., Lobocka, M., Moraru, C., Moreno Switt, A., Morozova, V., Nakavuma, J., Reyes Muñoz, A., Rūmnieks, J., Sarkar, B.L., Sullivan, M.B., Uchiyama, J., Wittmann, J., Yigang, T., Adriaenssens, E.M., 2023. Abolishment of morphology-based taxa and change to binomial species names: 2022 taxonomy update of the ICTV bacterial viruses subcommittee. Arch. Virol. 168, 74. https://doi.org/10.1007/s00705-022-05694-2.
UniProt Consortium, 2021. UniProt: the universal protein knowledgebase in 2021. Nucleic Acids Res. 49, D480–D489. https://doi.org/10.1093/nar/gkaa1100.
Ussery, D.W., 2002. DNA structure: A-, B- and Z-DNA helix families. eLS (Encyclopedia Life Sci.). https://doi.org/10.1038/npg.els.0003122.
Vafabakhsh, R., Ha, T., 2012. Extreme bendability of DNA less than 100 base pairs long revealed by single-molecule cyclization. Science 80 (337), 1097. https://doi.org/10.1126/science.1224139.
Wang, D.D., Yan, H., 2011. The relationship between periodic dinucleotides and the nucleosomal DNA deformation revealed by normal mode analysis. Phys. Biol. 8, 066004. https://doi.org/10.1088/1478-3975/8/6/066004.
Wing, R., Drew, H., Takano, T., Broka, C., Tanaka, S., Itakura, K., Dickerson, R.E., 1980. Crystal structure analysis of a complete turn of B-DNA. Nature 287, 755–758. https://doi.org/10.1038/287755a0.
Wu, Q., Zhou, W., Wang, J., Yan, H., 2011. Correlation between the flexibility and periodic dinucleotide patterns in yeast nucleosomal DNA sequences. J. Theor. Biol. 284, 92–98. https://doi.org/10.1016/j.jtbi.2011.06.026.
Xuan, G., Kong, J., Wang, Y., Lin, H., Wang, J., 2023. Characterization of the newly isolated Pseudomonas phage vB_Pae_LC3I3. Virus Res. 323, 198978. https://doi.org/10.1016/j.virusres.2022.198978.
Yamada, J.K., 2014. Resistance of Pseudomonas syringae pv. garcae to copper [Resistência de isolados de Pseudomonas syringae pv. garcae ao cobre]. Federal University of Lavras.
Yesudhas, D., Batool, M., Anwar, M.A., Panneerselvam, S., Choi, S., 2017. Proteins recognizing DNA: structural uniqueness and versatility of DNA-Binding domains in stem cell transcription factors. Genes 8, 192. https://doi.org/10.3390/genes8080192.
Yuan, Y., Peng, Q., Yang, S., Zhang, S., Fu, Y., Wu, Y., Gao, M., 2018. Isolation of A novel Bacillus thuringiensis phage representing A new phage lineage and characterization of its endolysin. Viruses 10, 611. https://doi.org/10.3390/v10110611.
Żaczek, M., Weber-Dabrowska, B., Górski, A., 2015. Phages in the global fruit and vegetable industry. J. Appl. Microbiol. 118, 537–556. https://doi.org/10.1111/jam.12700.
Zaika, S., Kharina, A., Chaika, D., Polyshchuk, V.P., 2013. Looking for killers: bacteriophages against phytopathogenic bacteria. Bull. Taras Shevchenko Natl. Univ. Kyiv 3, 86–89.
Zhang, Y., Basu, A., Ha, T., Bialek, W., 2020. Searching for sequence features that control DNA flexibility. arXiv:2012.06127v1 [physics.bio-ph]. https://doi.org/10.48550/arXiv.2012.06127.
Zhang, Y., Crothers, D.M., 2003. Statistical mechanics of sequence-dependent circular DNA and its application for DNA cyclization. Biophys. J. 84, 136–153. https://doi.org/10.1016/S0006-3495(03)74838-3.
Zhou, T., Shen, N., Yang, L., Abe, N., Horton, J., Mann, R.S., Bussemaker, H.J., Gordân, R., Rohs, R., 2015. Quantitative modeling of transcription factor binding specificities using DNA shape. Proc. Natl. Acad. Sci. USA 112, 4654–4659. https://doi.org/10.1073/pnas.1422023112.
Zhou, T., Yang, L., Lu, Y., Dror, I., Machado, A.C.D., Ghane, T., Di Felice, R., Rohs, R., 2013. DNAshape: a method for the high-throughput prediction of DNA structural features on a genomic scale. Nucleic Acids Res. 41, W56–W62. https://doi.org/10.1093/nar/gkt437.
Zhurkin, V.B., Lysov, Y.P., Ivanov, V.I., 1979. Anisotropic flexibility of DNA and the nucleosomal structure. Nucleic Acids Res. 6, 1081–1096. https://doi.org/10.1093/nar/6.3.1081.